Choice Theory

Individual decision-making under certainty

Objects of inquiry

Our study begins with individual decision-making under certainty
Items of interest include:

- Feasible set
- Objective function (Feasible set $\to \mathbb{R}$)
- Choice correspondence (Parameters $\rightrightarrows$ Feasible set)
- "Maximized" objective function (Parameters $\to \mathbb{R}$)

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We start with an even more general problem that only includes

- Feasible set
- Choice correspondence

A fairly innocent assumption will then allow us to treat this model as an optimization problem

Individual decision-making under certainty

Course outline

We will divide decision-making under certainty into three units:

1. Producer theory
   - Feasible set defined by technology
   - Objective function $p \cdot y$ depends on prices
2. Abstract choice theory
   - Feasible set totally general
   - Objective function may not even exist
3. Consumer theory
   - Feasible set defined by budget constraint and depends on prices
   - Objective function $u(x)$

Origins of rational choice theory

Choice theory aims to provide answers to

Positive questions Understanding how individual self-interest drives larger economic systems

Normative questions Objective criterion for utilitarian calculations

Values of the model

- Useful (somewhat)
  - Can often recover preferences from choices
  - Aligned with democratic values
  - But... interpersonal comparisons prove difficult
- Accurate (somewhat): many comparative statics results empirically verifiable
- Broad
  - Consumption and production
  - Lots of other things
- Compact
  - Extremely compact formulation
  - Ignores an array of other important "behavioral" factors

Simplifying assumptions

Very minimal:

1. Choices are made from some feasible set
2. Preferred things get chosen
3. Any pair of potential choices can be compared
4. Preferences are transitive
   (e.g., if apples are at least as good as bananas, and bananas are at least as good as cantaloupe, then apples are at least as good as cantaloupe)

Outline

- Preferences
  - Preference relations and rationality
  - From preferences to behavior
  - From behavior to preferences: "revealed preference"
- Utility functions
- Properties of preferences
- Behavioral critiques

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The set of all possible choices

We consider an entirely general set of possible choices

- Number of choices
  - Finite (e.g., types of drinks in my refrigerator)
  - Countably infinite (e.g., number of cars)
  - Uncountably infinite (e.g., amount of coffee)
  - Bounded *or* unbounded
- Order of choices
  - Fully ordered (e.g., years of schooling)
  - Partially ordered (e.g., AT&T cell phone plans)
  - Unordered (e.g., wives/husbands)

Note not all choices need be *feasible* in a particular situation

Preference relations

Definition (weak preference relation)

$\succsim$ is a binary relation on a set of possible choices X such that $x \succsim y$ iff "x is at least as good as y."

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Definition (strict preference relation)

$\succ$ is a binary relation on X such that $x \succ y$ ("x is strictly preferred to y") iff $x \succsim y$ but $y \not\succsim x$.

Definition (indifference)

$\sim$ is a binary relation on X such that $x \sim y$ ("the agent is indifferent between x and y") iff $x \succsim y$ and $y \succsim x$.

Properties of preference relations

Definition (completeness)

$\succsim$ on X is complete iff $\forall x, y \in X$, either $x \succsim y$ or $y \succsim x$.

Completeness implies that $x \succsim x$

Definition (transitivity)

$\succsim$ on X is transitive iff whenever $x \succsim y$ and $y \succsim z$, we have $x \succsim z$.

Rules out preference cycles except in the case of indifference

Definition (rationality)

$\succsim$ on X is rational iff it is both complete and transitive.

Summary of preference notation

	$y \succsim x$	$y \not\succsim x$
$x \succsim y$	$x \sim y$	$x \succ y$
$x \not\succsim y$	$y \succ x$	Ruled out by completeness

Summary of preference notation

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Can think of (complete) preferences as inducing a function

$$p\colon X \times X \to \{\succ, \sim, \prec\}$$

Other properties of rational preference relations

Assume $\succsim$ is rational. Then for all x, y, $z \in X$:

- Weak preference is reflexive: $x \succsim x$
- Indifference is
  - Reflexive: $x \sim x$
  - Transitive: $(x \sim y) \wedge (y \sim z) \implies x \sim z$
  - Symmetric: $x \sim y \iff y \sim x$
- Strict preference is
  - Irreflexive: $x \not\succ x$
  - Transitive: $(x \succ y) \wedge (y \succ z) \implies x \succ z$
- $(x \succ y) \wedge (y \succsim z) \implies x \succ z$, and
  $(x \succsim y) \wedge (y \succ z) \implies x \succ z$

Two strategies for modelling individual decision-making

1. Conventional approach
   Start from preferences, ask what choices are compatible

2. Revealed-preference approach
   Start from observed choices, ask what preferences are compatible
   - Can we test rational choice theory? How?
   - Are choices consistent with maximization of some objective function? Can we recover an objective function?
   - How can we use objective function—in particular, do interpersonal comparisons work? If so, how?

Choice rules

Definition (Choice rule)

Given preferences $\succsim$ over X, and choice set $B \subseteq X$, the choice rule is a correspondence giving the set of all "best" elements in B:

$$C(B, \succsim) \equiv \{x \in B : x \succsim y \text{ for all } y \in B\}.$$

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Theorem

Suppose $\succsim$ is complete and transitive and B finite and non-empty. Then $C(B, \succsim) \neq \varnothing$.

Proof of non-emptiness of choice correspondence

Proof.

Proof by mathematical induction on the number of elements in B.

Consider $|B| = 1$ so $B = \{x\}$; by completeness $x \succsim x$, so $x \in C(B, \succsim) \implies C(B, \succsim) \neq \varnothing$.

Suppose that for all $|B| = n \geq 1$, we have $C(B, \succsim) \neq \varnothing$. Consider A such that $|A| = n + 1$; thus $A = B \cup \{x\}$. We can consider some $y \in C(B, \succsim)$ by the inductive hypothesis. By completeness, either

1. $y \succsim x$, in which case $y \in C(A, \succsim)$.
2. $x \succsim y$, in which case $x \in C(A, \succsim)$ by transitivity.

Thus $C(A, \succsim) \neq \varnothing$

The inductive hypothesis holds for all finite n. □

Revealed preference

- Before, we used a known preference relation $\succsim$ to generate choice rule $C(\cdot, \succsim)$
- Now we suppose the agent reveals her preferences through her choices, which we observe; can we deduce a rational preference relation that could have generated them?

Definition (revealed preference choice rule)

Any $C_R : 2^X \to 2^X$ (where 2^X means the set of subsets of X) such that for all $A \subseteq X$, we have $C_R(A) \subseteq A$.

If $C_R(\cdot)$ could be generated by a rational preference relation (i.e., there exists some complete, transitive $\succsim$ such that $C_R(A) = C(A, \succsim)$ for all A), we say it is rationalizable

Examples of revealed preference choice rules

Suppose we know $C_R(\cdot)$ for

- $A \equiv \{a, b\}$
- $B \equiv \{a, b, c\}$

$C_R(\{a, b\})$	$C_R(\{a, b, c\})$	Possibly rationalizable?
$\{a\}$	$\{c\}$	
$\{a\}$	$\{a\}$	
$\{a, b\}$	$\{c\}$	

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A necessary condition for rationalizability

Suppose that $C_R(\cdot)$ is rationalizable (in particular, it is generated by $\succsim$), and we observe $C_R(A)$ for some $A \subseteq X$ such that

- $a \in C_R(A)$ (a was chosen $\iff a \succsim z$ for all $z \in A$)
- $b \in A$ (b could have been chosen)

We can infer that $a \succsim b$

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Now consider some $B \subseteq X$ such that

- $a \in B$
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We can infer that $b \succsim a$

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Thus $a \sim b$, hence $a \in C_R(B)$ and $b \in C_R(A)$ by transitivity

Houthaker's Axiom of Revealed Preferences

A rationalizable choice rule $C_R(\cdot)$ must therefore satisfy "HARP":

Definition (Houthaker's Axiom of Revealed Preferences)

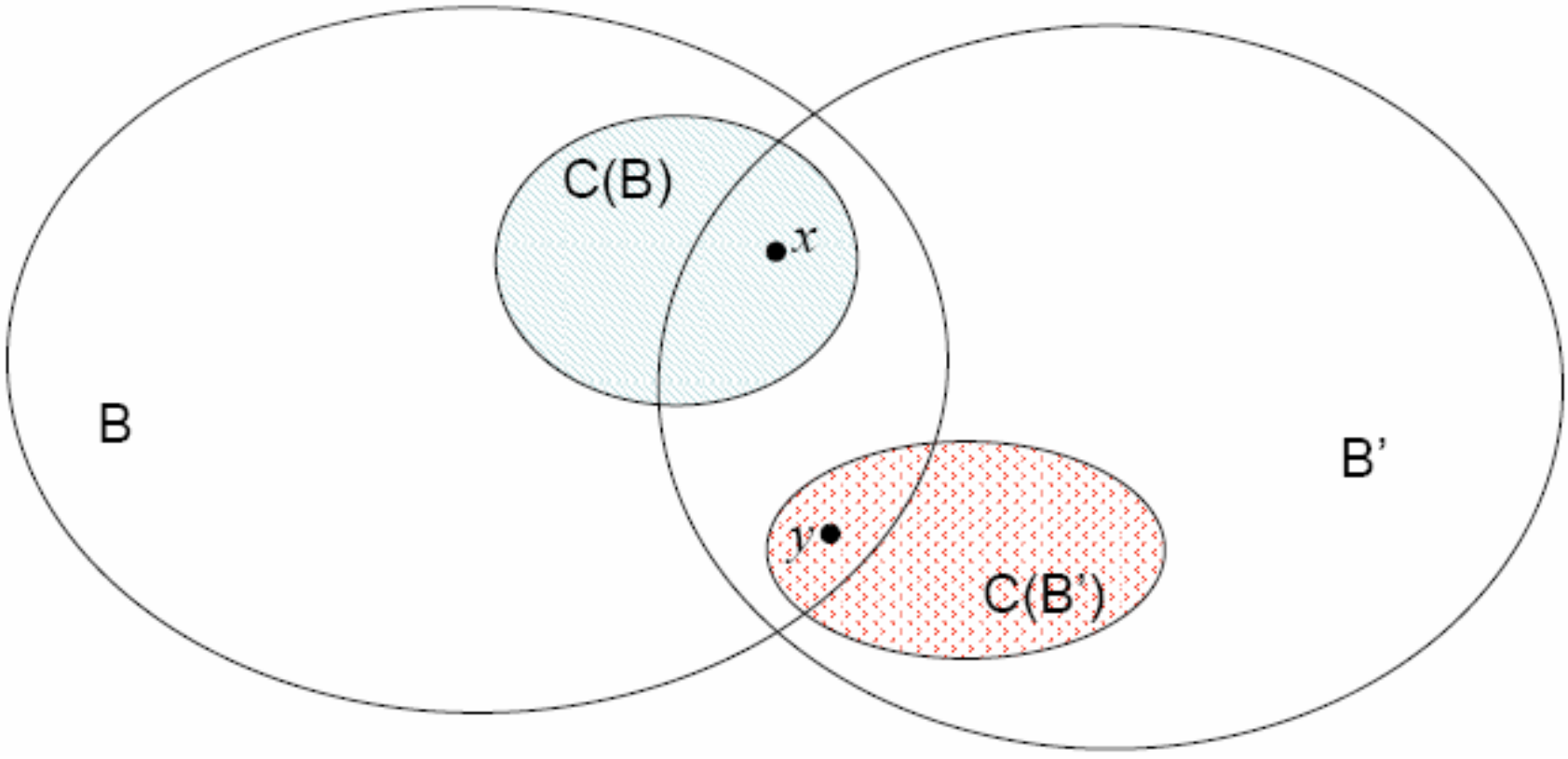
Revealed preferences $C_R : 2^X \to 2^X$ satisfies HARP iff $\forall a, b \in X$ and $\forall A, B \subseteq X$ such that

- $\{a, b\} \subseteq A$ and $a \in C_R(A)$; and
- $\{a, b\} \subseteq B$ and $b \in C_R(B)$,

we have that $a \in C_R(B)$ (and $b \in C_R(A)$).

Illustrating HARP

A violation of HARP:



Example of HARP

Suppose

1. Revealed preferences $C_R(\cdot)$ satisfy HARP, and that
2. $C_R(\cdot)$ is nonempty-valued (except for $C_R(\varnothing)$)

- If $C_R(\{a, b\}) = \{b\}$, what can we conclude about $C_R(\{a, b, c\})$?

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$$C_R(\{a,b,c\}) \in \big\{\{b\},\{c\},\{b,c\}\big\}$$

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- If $C_R(\{a,b,c\}) = \{b\}$, what can we conclude about $C_R(\{a,b\})$?

$$C_R(\{a,b\}) = \{b\}$$

HARP is necessary and sufficient for rationalizability I

Theorem

Suppose revealed preference choice rule $C_R: 2^X \to 2^X$ *is nonempty-valued (except for* $C_R(\varnothing)$*). Then* $C_R(\cdot)$ *satisfies HARP iff there exists a rational preference relation* $\succsim$ *such that* $C_R(\cdot) = C(\cdot, \succsim)$.

Proof.

Rationalizability $\Rightarrow$ HARP as argued above.

Rationalizability $\Leftarrow$ HARP: suppose $C_R(\cdot)$ satisfies HARP, we will construct a "revealed preference relation" $\succsim_c$ that generates $C_R(\cdot)$.

For any x and y, let $x \succsim_c y$ iff there exists some $A \subseteq X$ such that $y \in A$ and $x \in C_R(A)$.

We must show that $\succsim_c$ is complete, transitive, and generates C (i.e., $C_R(\cdot) = C(\cdot, \succsim_c)$).

HARP is necessary and sufficient for rationalizability II

Proof (continued).

1. $C_R(\cdot)$ is nonempty-valued, so either $x \in C_R\big(\{x,y\}\big)$ or $y \in C_R\big(\{x,y\}\big)$. Thus either $x \succsim_c y$ or $y \succsim_c x$.
2. Suppose $x \succsim_c y \succsim_c z$ and consider $C_R\big(\{x,y,z\}\big) \neq \varnothing$. Thus one (or more) of the following must hold:
   1. $x \in C_R\big(\{x,y,z\}\big) \implies x \succsim_c z$.
   2. $y \in C_R\big(\{x,y,z\}\big)$.
      But $x \succsim_c y$, so by HARP $x \in C_R\big(\{x,y,z\}\big) \implies x \succsim_c z$ by 1.
   3. $z \in C_R\big(\{x,y,z\}\big)$.
      But $y \succsim_c z$, so by HARP $y \in C_R\big(\{x,y,z\}\big) \implies x \succsim_c z$ by 2.
3. We must show that $x \in C_R(B)$ iff $x \succsim_c y$ for all $y \in B$.
   - $x \in C_R(B) \implies x \succsim_c y$ by construction of $\succsim_c$.
   - By nonempty-valuedness, there must be some $y \in C_R(B)$; by HARP, $x \succsim_c y$ implies that $x \in C_R(B)$. □

Revealed preference and limited data

Our discussion relies on all preferences being observed

- *All* elements of $C_R(A)$
- $C_R(A)$ for *every* $A \subseteq X$

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Our discussion relies on all preferences being observed. . . real data is typically more limited

- *All* elements of $C_R(A)$. . . we may only see *one* element of A i.e., $\widetilde{C}_R(A) \in C_R(A)$
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  i.e., $\widehat{C}_R(A) \colon \mathcal{B} \to 2^X$ for $\mathcal{B} \subset 2^X$ with $\widehat{C}_R(A) = C_R(A)$

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Other "axioms of revealed preference" hold in these environments

- Weak Axiom of Revealed Preference (WARP)
- Generalized Axiom of Revealed Preference (GARP)—necessary and sufficient condition for rationalizability

Weak Axiom of Revealed Preferences I

Definition (Weak Axiom of Revealed Preferences)

Revealed preferences $\widehat{C}_R : \mathcal{B} \to 2^X$ defined only for choice sets $\mathcal{B} \subseteq 2^X$ satisfies WARP iff $\forall a, b \in X$ and $\forall A, B \in \mathcal{B}$ such that

- $\{a, b\} \subseteq A$ and $a \in \widehat{C}_R(A)$; and
- $\{a, b\} \subseteq B$ and $b \in \widehat{C}_R(B)$,

we have that $a \in \widehat{C}_R(B)$ (and $b \in \widehat{C}_R(A)$).

HARP is WARP with all possible choice sets (i.e,. $\mathcal{B} = 2^X$)

WARP is necessary but not sufficient for rationalizability

Weak Axiom of Revealed Preferences II

WARP is not sufficient for rationalizability

Example

Consider $\widehat{C}_R \colon \mathcal{B} \to 2^{\{a,b,c\}}$ defined for choice sets $\mathcal{B} \equiv \big\{\{a,b\},\{b,c\},\{c,a\}\big\} \subseteq 2^{\{a,b,c\}}$ with:

- $\widehat{C}_R\big(\{a,b\}\big) = \{a\}$,
- $\widehat{C}_R\big(\{b,c\}\big) = \{b\}$, and
- $\widehat{C}_R\big(\{c,a\}\big) = \{c\}$.

$\widehat{C}_R(\cdot)$ satisfies WARP, but is not rationalizable.

Think of $\widehat{C}_R(\cdot)$ as a restriction of some $C_R \colon 2^{\{a,b,c\}} \to 2^{\{a,b,c\}}$; there is no $C_R\big(\{a,b,c\}\big)$ consistent with HARP

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- Properties of preferences
- Behavioral critiques

From abstract preferences to maximization

- Our model of choice so far is entirely abstract
- Utility assigns a numerical ranking to each possible choice
- By assigning a utility to each element of X, we turn the choice problem into an optimization problem

Definition (utility function)

Utility function $u: X \to \mathbb{R}$ represents $\succsim$ on X iff for all $x, y \in X$,

$$x \succsim y \iff u(x) \geq u(y).$$

Then the choice rule is

$$C(B, \succsim) \equiv \{x \in B : x \succsim y \text{ for all } y \in B\} = \operatorname*{argmax}_{x \in B} u(x)$$

Utility representation implies rationality

Theorem

If utility function $u\colon X \to \mathbb{R}$ represents $\succsim$ on X, then $\succsim$ is rational.

Proof.

For any x, $y \in X$, we have $u(x)$, $u(y) \in \mathbb{R}$, so either $u(x) \geq u(y)$ or $u(y) \geq u(x)$. Since $u(\cdot)$ represents $\succsim$, either $x \succsim y$ or $y \succsim x$; i.e., $\succsim$ is complete.

Suppose $x \succsim y \succsim z$. Since $u(\cdot)$ represents $\succsim$, we know $u(x) \geq u(y) \geq u(z)$. Thus $u(x) \geq u(z) \implies x \succsim z$. $\succsim$ is transitive. $\square$

Ordinality of utility and interpersonal comparisons

Note that $\succsim$ is represented by any function satisfying

$$x \succsim y \iff u(x) \geq u(y)$$

for all $x, y \in X$

Thus any increasing monotone transformation of $u(\cdot)$ also represents $\succsim$

- The property of representing $\succsim$ is ordinal
- There is no such thing as a "util"

Failure of interpersonal comparisons

Interpersonal comparisons are impossible using this theory

1. Disappointing to original utilitarian agenda
2. Rawls (following Kant, following...) attempts to solve this by asking us to consider only a single chooser
3. "Just noticeable difference" suggests defining one util as the smallest difference an individual can notice
   - $x \succsim y$ iff $u(x) \geq u(y) - 1$
   - Note $\succ$ is transitive, but $\succsim$ is not

Can we find a utility function representing $\succsim$? I

Theorem

Any complete and transitive preference relation $\succsim$ on a finite set X can be represented by some utility function $u\colon X \to \{1, \ldots, n\}$ where $n \equiv |X|$.

Intuitive argument:

1. Assign the "top" elements of X utility $n = |X|$
2. Discard them; we are left with a set X'
3. If $X' = \varnothing$, we are done; otherwise return to step 1 with the set X'

Can we find a utility function representing $\succsim$? II

Proof.

Proof by mathematical induction on $n \equiv |X|$. The theorem holds trivially for $n = 0$, since $X = \varnothing$.

Suppose the theorem holds for sets with at most n elements. Consider a set X with $n+1$ elements. $C(X, \succsim) \neq \varnothing$, so $Y \equiv X \setminus C(X, \succsim)$ has at most n elements. By inductive hypothesis, preferences on Y are represented by some $u\colon Y \to \{1, \ldots, n\}$. We extend u to X by setting $u(x) = n+1$ for all $x \in C(X, \succsim)$.

We must show that this extended u represents $\succsim$ on X.

Can we find a utility function representing $\succsim$? III

Proof (continued).

We must show that this extended u represents $\succsim$ on X; i.e., that for all x and $y \in X$, we have $x \succsim y$ iff $u(x) \geq u(y)$.

- $x \succsim y \Longrightarrow u(x) \geq u(y)$. If $x \in C(X, \succsim)$, then $u(x) = n + 1 \geq u(y)$. If $x \notin C(X, \succsim)$, then by transitivity $y \notin C(X, \succsim)$, so x and $y \in Y$. Since u represents $\succsim$ on Y, we must have $u(x) \geq u(y)$.
- $x \succsim y \Longleftarrow u(x) \geq u(y)$. If $u(x) = n + 1$, then $x \in C(X, \succsim)$, hence $x \succsim y$. If $u(x) \leq n$, then x and $y \in Y$. Since u represents $\succsim$ on Y, we must have $x \succsim y$.

Thus the inductive hypothesis holds for all finite n. □

What if $|X| = \infty$? I

If X is infinite, our proof doesn't go through, but we still *may* be able to represent $\succsim$ by a utility function

Example

Preferences over $\mathbb{R}_+$ with $x_1 \succsim x_2$ iff $x_1 \geq x_2$.

$\succsim$ can be represented by $u(x) = x$. (It can also be represented by other utility functions.)

What if $|X| = \infty$? II

However, if X is infinite we can't necessarily represent $\succsim$ by a utility function

Example (lexicographic preferences)

Preferences over $[0,1]^2 \subseteq \mathbb{R}^2$ with $(x_1, y_1) \succsim (x_2, y_2)$ iff

- $x_1 > x_2$, or
- $x_1 = x_2$ and $y_1 \geq y_2$.

Lexicographic preferences can't be represented by a utility function

- There are no indifference curves
- A utility function would have to be an order-preserving one-to-one mapping from the unit square to the real line (both are infinite, but they are "different infinities")

Continuous preferences I

Definition (continuous preference relation)

A preference relation $\succsim$ on X is continuous iff for any sequence $\{(x_n, y_n)\}_{n=1}^{\infty}$ with $x_n \succsim y_n$ for all n,

$$\lim_{n\to\infty} x_n \succsim \lim_{n\to\infty} y_n.$$

Equivalently, $\succsim$ is continuous iff for all $x \in X$, the upper and lower contour sets of x

$$\text{UCS}(x) \equiv \{\xi \in X : \xi \succsim x\}$$
$$\text{LCS}(x) \equiv \{\xi \in X : x \succsim \xi\}$$

are both closed sets.

Continuous preferences II

Theorem

A continuous, rational preference relation $\succsim$ on $X \subseteq \mathbb{R}^n$ can be represented by a continuous utility function $u\colon X \to \mathbb{R}$.

(Note it may also be represented by noncontinuous utility functions)

Full proof in Debron and MWG; abbreviated proof in notes

Outline

- Preferences
  - Preference relations and rationality
  - From preferences to behavior
  - From behavior to preferences: "revealed preference"
- Utility functions
- Properties of preferences
- Behavioral critiques

Reasons for restricting preferences

Analytical tractability often demands restricting "allowable" preferences

- Some restrictions are mathematical conveniences and cannot be empirically falsified (e.g., continuity)
- Some hold broadly (e.g., monotonicity)
- Some require situational justification

Restrictions on preferences imply restrictions on utility functions

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Restrictions on preferences imply restrictions on utility functions

Assumptions for the rest of this section

1. $\succsim$ is rational (i.e., complete and transitive).
2. For simplicity, we assume preferences over $X \subseteq \mathbb{R}^n$.

Properties of rational $\succsim$: Nonsatiation

Definition (monotonicity)

$\succsim$ is monotone iff $x > y \implies x \succsim y$. (N.B. MWG differs)
$\succsim$ is strictly monotone iff $x > y \implies x \succ y$.

i.e., more of something is (strictly) better

Definition (local non-satiation)

$\succsim$ is locally non-satiated iff for any y and $\varepsilon > 0$, there exists x such that $\|x - y\| \leq \varepsilon$ and $x \succ y$.

implies there are no "thick" indifference curves
$\succsim$ is locally non-satiated iff $u(\cdot)$ has no local maxima in X

Properties of rational $\succsim$: Convexity I

Convex preferences capture the idea that agents like diversity

1. Satisfying in some ways: rather alternate between juice and soda than have either one every day
2. Unsatisfying in others: rather have a glass of either one than a mixture
3. Key question is granularity of goods aggregation
   - Over time? What period?
   - Over what "bite size"?

Properties of rational $\succsim$: Convexity II

Definition (convexity)

$\succsim$ is convex iff $x \succsim y$ and $x' \succsim y$ together imply that

$$tx + (1-t)x' \succsim y \text{ for all } t \in (0,1).$$

Equivalently, $\succsim$ is convex iff the upper contour set of any y (i.e., $\{x \in X : x \succsim y\}$) is a convex set.

$\succsim$ is strictly convex iff $x \succsim y$ and $x' \succsim y$ (with $x \neq x'$) together imply that

$$tx + (1-t)x' \succ y \text{ for all } t \in (0,1).$$

i.e., one never gets worse off by mixing goods

$\succsim$ is (strictly) convex iff $u(\cdot)$ is (strictly) quasiconcave ▸ Definition

Implications for utility representation

Property of $\succsim$	Property of $u(\cdot)$
Monotone	Nondecreasing
Strictly monotone	Increasing
Locally non-satiated	Has no local maxima in X
Convex	Quasiconcave
Strictly convex	Strictly quasiconcave

Properties of rational $\succsim$: Homotheticity

Definition (homotheticity)

$\succsim$ is homothetic iff for all x, y, and all $\lambda > 0$,

$$x \succsim y \iff \lambda x \succsim \lambda y.$$

Continuous, strictly monotone $\succsim$ is homothetic iff it can be represented by a utility function that is

Properties of rational $\succsim$: Homotheticity

Definition (homotheticity)

$\succsim$ is homothetic iff for all x, y, and all $\lambda > 0$,

$$x \succsim y \iff \lambda x \succsim \lambda y.$$

Continuous, strictly monotone $\succsim$ is homothetic iff it can be represented by a utility function that is homogeneous of degree one (note it can also be represented by utility functions that aren't)

Properties of rational $\succsim$: Separability I

Suppose rational $\succsim$ over $X \times Y \subseteq \mathbb{R}^{p+q}$

- First p goods form some "group" $x \in X \subseteq \mathbb{R}^p$
- Other goods $y \in Y \subseteq \mathbb{R}^q$

Separable preferences

"Preferences over X do not depend on y" means that

$$(x', y_1) \succsim (x, y_1) \iff (x', y_2) \succsim (x, y_2)$$

for all x, $x' \in X$ and all y_1, $y_2 \in Y$.
Note the definition is *not* symmetric in X and Y.

The critical assumption for empirical analysis of preferences

Properties of rational $\succsim$: Separability II

Example

$X = \{\text{wine}, \text{beer}\}$ and $Y = \{\text{cheese}, \text{pretzels}\}$ with strict preference ranking

1. $(\text{wine}, \text{cheese}) \succ$
2. $(\text{wine}, \text{pretzels}) \succ$
3. $(\text{beer}, \text{pretzels}) \succ$
4. $(\text{beer}, \text{cheese})$.

Utility representation of separable preferences: theorem

Theorem

Suppose $\succsim$ *on* $X \times Y$ *is represented by* $u(x, y)$*. Then preferences over* X *do not depend on* y *iff there exist functions* $v : X \to \mathbb{R}$ *and* $U : \mathbb{R} \times Y \to \mathbb{R}$ *such that*

1. $U(\cdot, \cdot)$ *is increasing in its first argument, and*
2. $u(x, y) = U\big(v(x), y\big)$ *for all* (x, y)*.*

Utility representation of separable preferences: example

Example

Preferences over beverages do not depend on your snack, and are represented by $u(\cdot, \cdot)$, where

$$\begin{aligned} u(\text{wine}, \text{cheese}) &= 4 \\ u(\text{wine}, \text{pretzels}) &= 3 \\ u(\text{beer}, \text{pretzels}) &= 2 \\ u(\text{beer}, \text{cheese}) &= 1. \end{aligned}$$

Utility representation of separable preferences: example

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Preferences over beverages do not depend on your snack, and are represented by $u(\cdot, \cdot)$, where

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Let $v(\text{wine}) \equiv 3$ and $v(\text{beer}) \equiv 2$.

Utility representation of separable preferences: example

Example

Preferences over beverages do not depend on your snack, and are represented by $u(\cdot, \cdot)$, where

$$\begin{array}{lll} u(\text{wine}, \text{cheese}) & = 4 & \text{And let} \quad U(3, \text{cheese}) \equiv 4 \\ u(\text{wine}, \text{pretzels}) & = 3 & \quad\quad\;\; U(3, \text{pretzels}) \equiv 3 \\ u(\text{beer}, \text{pretzels}) & = 2 & \quad\quad\;\; U(2, \text{pretzels}) \equiv 2 \\ u(\text{beer}, \text{cheese}) & = 1. & \quad\quad\;\; U(2, \text{cheese}) \equiv 1. \end{array}$$

Let $v(\text{wine}) \equiv 3$ and $v(\text{beer}) \equiv 2$.

Utility representation of separable preferences: example

Example

Preferences over beverages do not depend on your snack, and are represented by $u(\cdot,\cdot)$, where

$$\begin{array}{lll} u(\text{wine},\text{cheese}) & = 4 & \text{And let} \quad U(3,\text{cheese}) \equiv 4 \\ u(\text{wine},\text{pretzels}) & = 3 & \quad\quad\;\; U(3,\text{pretzels}) \equiv 3 \\ u(\text{beer},\text{pretzels}) & = 2 & \quad\quad\;\; U(2,\text{pretzels}) \equiv 2 \\ u(\text{beer},\text{cheese}) & = 1. & \quad\quad\;\; U(2,\text{cheese}) \equiv 1. \end{array}$$

Let $v(\text{wine}) \equiv 3$ and $v(\text{beer}) \equiv 2$.

Thus

1. $U(\cdot,\cdot)$ is increasing in its first argument, and
2. $u(x,y) = U\big(v(x),y\big)$ for all (x,y).

Utility representation of separable preferences: proof I

Proof.

Conditions $\Longrightarrow$ separability: If $u(x, y) = U\big(v(x), y\big)$ with $U(\cdot, \cdot)$ increasing in its first argument, then preferences over X given any y are represented by $v(x)$ and do not depend on y.

Conditions $\Longleftarrow$ separability: We assume preferences over X do not depend on y, construct a U and v, and then show that they satisfy

1. $u(x, y) = U\big(v(x), y\big)$ for all (x, y), and
2. $U(\cdot, \cdot)$ is increasing in its first argument.

Utility representation of separable preferences: proof II

Proof (continued).

- Fix some $y_0 \in Y$, and let $v(x) \equiv u(x, y_0)$.
- Consider every α in the range of $v(\cdot)$; that is there is (at least one) $v^{-1}(\alpha)$ such that $v\left(v^{-1}(\alpha)\right) = \alpha$. Define

$$U(\alpha, y) \equiv u\left(v^{-1}(\alpha), y\right). \tag{1}$$

Note that

$$\begin{aligned} u\left(v^{-1}\left(v(x)\right), y_0\right) = v\left(v^{-1}\left(v(x)\right)\right) = v(x) = u(x, y_0) & \\ \left(v^{-1}\left(v(x)\right), y_0\right) \sim & \quad (x, y_0) \\ \left(v^{-1}\left(v(x)\right), y\right) \sim & \quad (x, y). \end{aligned} \tag{2}$$

Utility representation of separable preferences: proof III

Proof (continued).

By 1 and 2,

$$U\big(v(x), y\big) = u\big(v^{-1}\big(v(x)\big), y\big) = u(x, y).$$

Choose any $y \in Y$ and any x, $x' \in X$ such that $v(x') > v(x)$:

$$\begin{aligned} u(x', y_0) > u(x, y_0) \Longrightarrow (x', y_0) &\succ (x, y_0) \\ (x', y) &\succ (x, y) \\ u(x', y) &> u(x, y) \\ U\big(v(x'), y\big) &> U\big(v(x), y\big) \end{aligned}$$

so $U(\cdot, \cdot)$ is increasing in its first argument for all y. □

Properties of rational $\succsim$: Quasi-linearity I

Suppose rational $\succsim$ over $X \equiv \mathbb{R} \times Y$

- First good is the numeraire (a.k.a. "good zero" or "good one," confusingly): think money
- Other goods general; need not be in $\mathbb{R}^n$

Theorem

Suppose rational $\succsim$ on $X \equiv \mathbb{R} \times Y$ satisfies the "numeraire properties":

1. *Good 1 is valuable*: $(t, y) \succsim (t', y) \iff t \geq t'$ *for all* y;
2. *Compensation is possible*: *For every* y, $y' \in Y$, *there exists some* $t \in \mathbb{R}$ *such that* $(0, y) \sim (t, y')$;
3. *No wealth effects*: *If* $(t, y) \succsim (t', y')$, *then for all* $d \in \mathbb{R}$, $(t + d, y) \succsim (t' + d, y')$.

...

Properties of rational $\succsim$: Quasi-linearity II

Theorem (continued.)

Then there exists a utility function representing $\succsim$ of the form $u(t, y) = t + v(y)$ for some $v : Y \to \mathbb{R}$. (Note it can also be represented by utility functions that aren't of this form.)

Conversely, any $\succsim$ on $X = \mathbb{R} \times Y$ represented by a utility function of the form $u(t, y) = t + v(y)$ satisfies the above properties.

Properties of rational $\succsim$: Quasi-linearity III

Proof.

Suppose the numeraire properties hold. Fix some $\bar{y} \in Y$. Define a function $v\colon Y \to \mathbb{R}$ such that $(0, y) \sim \big(v(y), \bar{y}\big)$; this is possible by condition 2.

By condition 3, for any (t, y) and (t', y'), we have $(t, y) \sim \big(t + v(y), \bar{y}\big)$ and $(t', y') \sim \big(t' + v(y'), \bar{y}\big)$. Thus $(t, y) \succsim (t', y')$ iff $\big(t + v(y), \bar{y}\big) \succsim \big(t' + v(y'), \bar{y}\big)$ (by transitivity), which holds by condition 1 iff $t + v(y) \geq t' + v(y')$.

The converse is trivial. □

Outline

- Preferences
  - Preference relations and rationality
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Problems with rational choice

Rational choice theory plays a central role in most tools of economic analysis

But. . . significant research calls into question underlying assumptions, identifying and explaining deviations using

- Psychology
- Sociology
- Cognitive neuroscience ("neuroeconomics")

Context-dependent choice

Choices appear to be highly situational, depending on

1. Other available options
2. Way that options are "framed"
3. Social context/emotional state

Numerous research projects consider these effects in real-world and laboratory settings

Non-considered choice

Rational choice theory depends on a *considered* comparison of options

- Pairwise comparison
- Utility maximization

Many actual choices appear to be made using

1. Intuitive reasoning
2. Heuristics
3. Instinctive desire

Consumer Theory

Individual decision-making under certainty

Course outline

We will divide decision-making under certainty into three units:

1. Producer theory
   - Feasible set defined by technology
   - Objective function $p \cdot y$ depends on prices
2. Abstract choice theory
   - Feasible set totally general
   - Objective function may not even exist
3. Consumer theory
   - Feasible set defined by budget constraint and depends on prices
   - Objective function $u(x)$

The consumer problem

Utility Maximization Problem

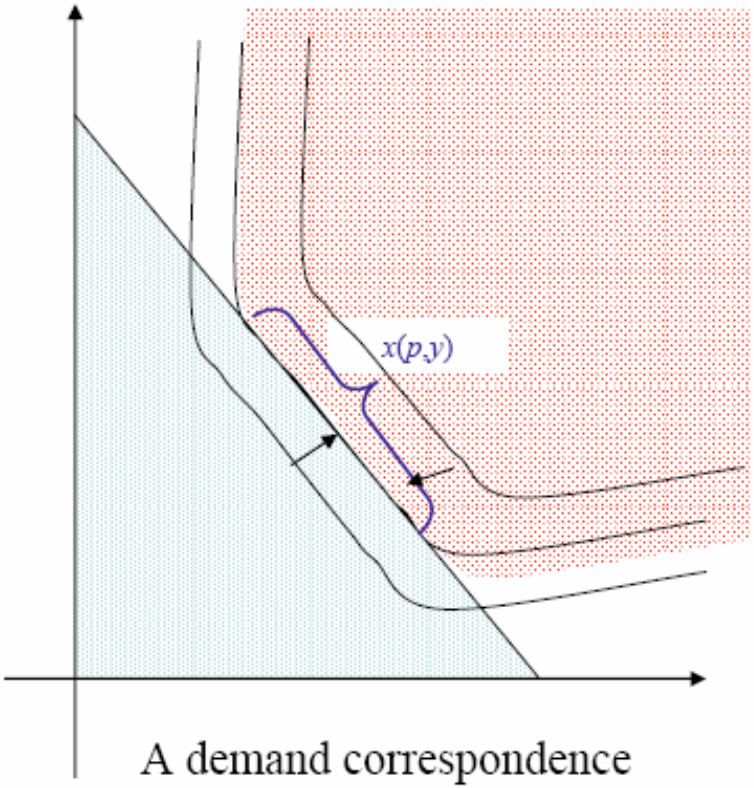
$$\max_{x \in \mathbb{R}^n_+} u(x) \text{ such that } \underbrace{p \cdot x}_{\text{Expenses}} \leq w$$

where p are the prices of goods and w is the consumer's "wealth."

This type of choice set is a budget set

$$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$$

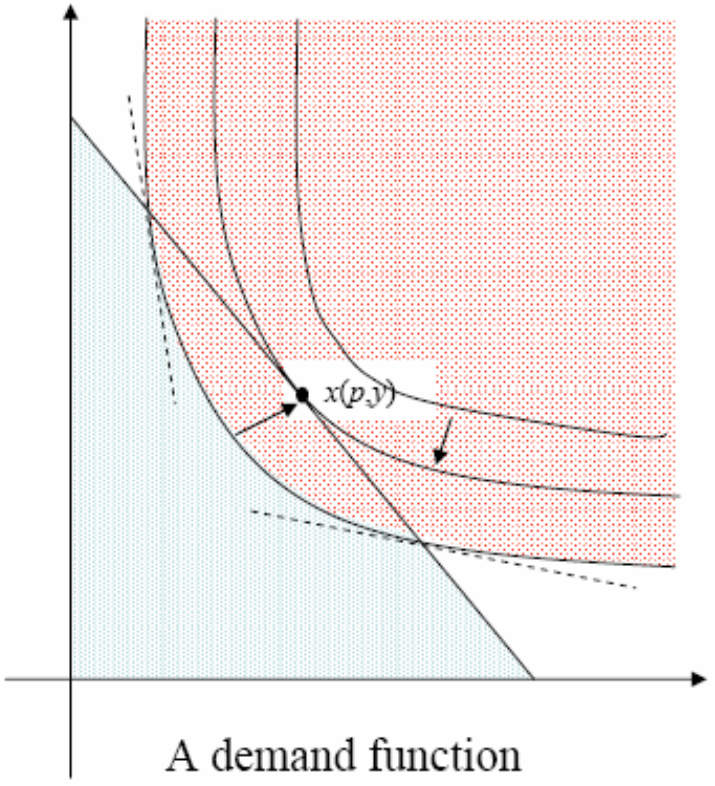
Illustrating the Utility Maximization Problem



A demand correspondence

↓

(Convex $\succsim$)



A demand function
(a single valued set)

↓

(Strictly convex $\succsim$)

Assumptions underlying the UMP

Note that

- Utility function is general (but assumed to exist—a restriction of preferences)
- Choice set defined by linear budget constraint
  - Consumers are price takers
  - Prices are linear
  - Perfect information: prices are all known
- Finite number of goods
  - Goods are described by quantity and price
  - Goods are divisible
  - Goods may be time- or situation-dependent
  - Perfect information: goods are all well understood

Outline

- The utility maximization problem
  - Marshallian demand and indirect utility
  - First-order conditions of the UMP
  - Recovering demand from indirect utility
- The expenditure minimization problem
- Wealth and substitution effects
  - The Slutsky equation
  - Comparative statics properties

Outline

- The utility maximization problem
  - Marshallian demand and indirect utility
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Utility maximization problem

The consumer's Marshallian demand is given by correspondence $x\colon \mathbb{R}^n \times \mathbb{R} \rightrightarrows \mathbb{R}^n_+$

$$x(p,w) \equiv \underset{x\in\mathbb{R}^n_+\colon p\cdot x\leq w}{\operatorname{argmax}} u(x) \equiv \underset{x\in B(p,w)}{\operatorname{argmax}} u(x)$$
$$= \left\{x \in \mathbb{R}^n_+ \colon p\cdot x \leq w \text{ and } u(x) = v(p,w)\right\}$$

Resulting indirect utility function is given by

$$v(p,w) \equiv \sup_{x\in\mathbb{R}^n_+\colon p\cdot x\leq w} u(x) \equiv \sup_{x\in B(p,w)} u(x)$$

Properties of Marshallian demand and indirect utility

Theorem

$v(p, w)$ and $x(p, w)$ are homogeneous of degree zero. That is, for all p, w, and $\lambda > 0$,

$$v(\lambda p, \lambda w) = v(p, w) \text{ and } x(\lambda p, \lambda w) = x(p, w).$$

These are "no money illusion" conditions

Proof.

$B(\lambda p, \lambda w) = B(p, w)$, so consumers are solving the same problem. □

Implications of restrictions on preferences: continuity

Theorem

If preferences are continuous, $x(p, w) \neq \varnothing$ for every $p \gg \mathbf{0}$ and $w \geq 0$.

i.e., Consumers choose *something*

Proof.

$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$ is a closed, bounded set. Continuous preferences can be represented by a continuous utility function $\tilde{u}(\cdot)$, and a continuous function achieves a maximum somewhere on a closed, bounded set. Since $\tilde{u}(\cdot)$ represents the same preferences as $u(\cdot)$, we know $\tilde{u}(\cdot)$ must achieve a maximum precisely where $u(\cdot)$ does. □

Implications of restrictions on preferences: convexity I

Theorem

If preferences are convex, then $x(p, w)$ is a convex set for every $p \gg \mathbf{0}$ and $w \geq 0$.

Proof.

$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$ is a convex set.
If x, $x' \in x(p, w)$, then $x \sim x'$.
For all $\lambda \in [0, 1]$, we have $\lambda x + (1 - \lambda)x' \in B(p, w)$ by convexity of $B(p, w)$ and $\lambda x + (1 - \lambda)x' \succsim x$ by convexity of preferences. Thus

$$\lambda x + (1 - \lambda)x' \in x(p, w).$$

□

Implications of restrictions on preferences: convexity II

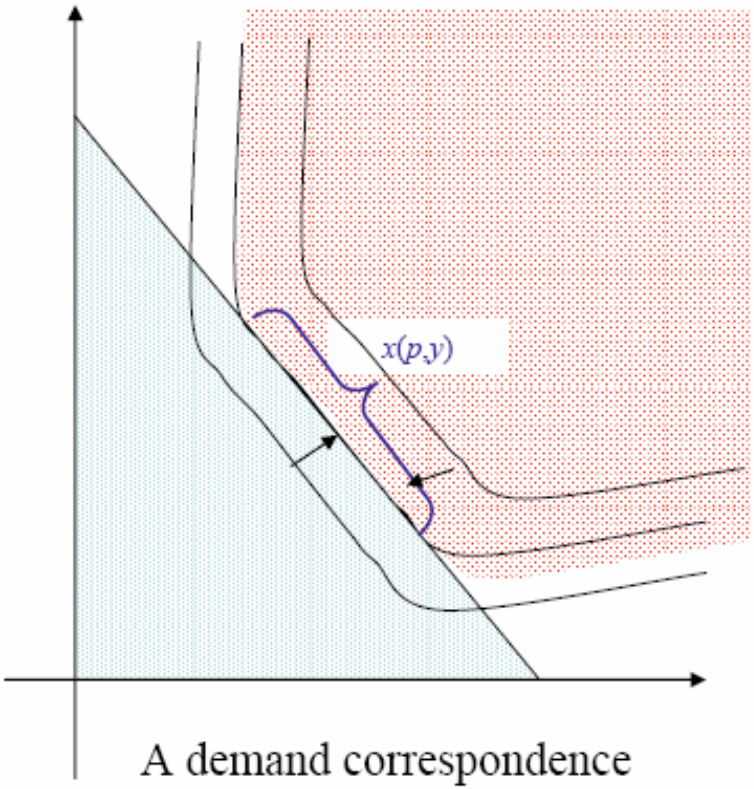
Theorem

If preferences are strictly convex, then $x(p, w)$ is single-valued for every $p \gg \mathbf{0}$ and $w \geq 0$.

Proof.

$B(p, w) \equiv \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}$ is a convex set.
If x, $x' \in x(p, w)$, then $x \sim x'$. Suppose $x \neq x'$.
For all $\lambda \in (0, 1)$, we have $\lambda x + (1 - \lambda)x' \in B(p, w)$ by convexity of $B(p, w)$ and $\lambda x + (1 - \lambda)x' \succ x$ by convexity of preferences.
But this contradicts the fact that $x \in x(p, w)$. Thus $x = x'$. □

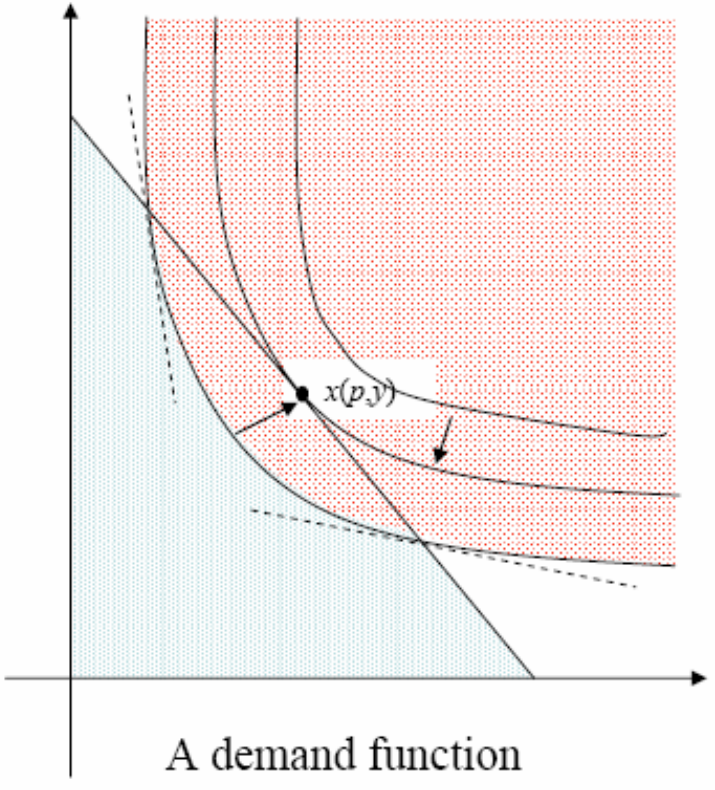
Implications of restrictions on preferences: convexity III



A demand correspondence

↓

(Convex $\succsim$)



A demand function
(a single valued set)

↓

(Strictly convex $\succsim$)

Implications of restrictions on preferences: non-satiation I

Definition (Walras' Law)

$p \cdot x = w$ for every $p \gg \mathbf{0}$, $w \geq 0$, and $x \in x(p, w)$.

Theorem

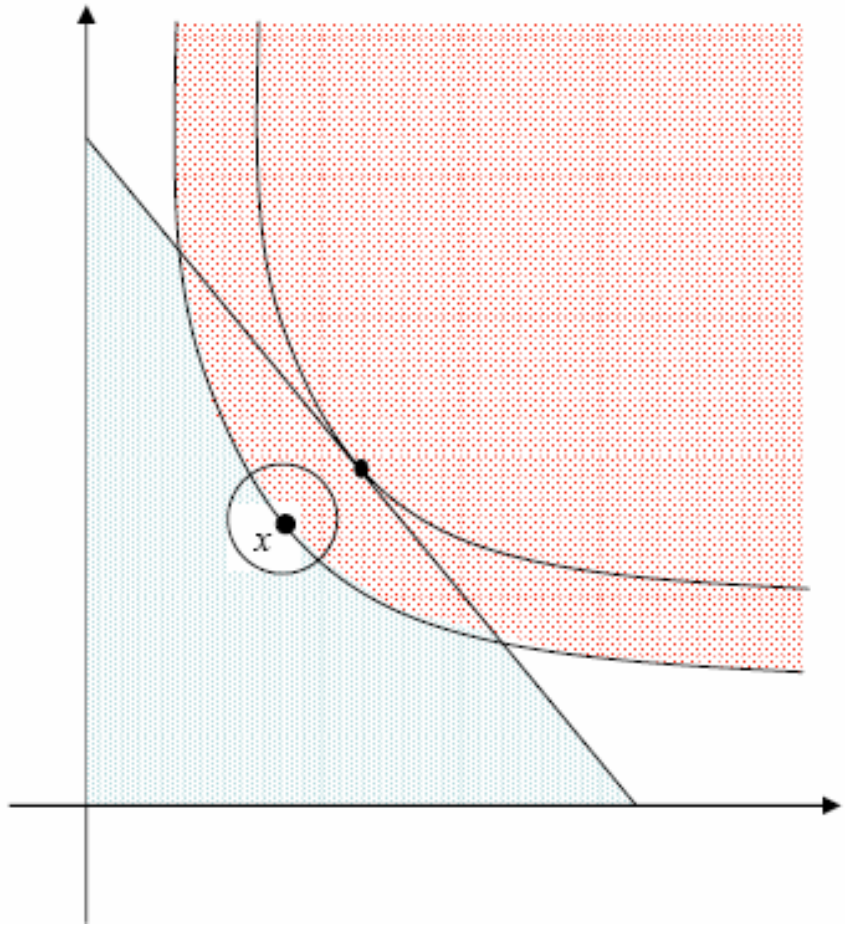
If preferences are locally non-satiated, then Walras' Law holds.

This allows us to replace the inequality constraint in the UMP with an equality constraint

Implications of restrictions on preferences: non-satiation II

Proof.

Suppose that $p \cdot x < w$ for some $x \in x(p, w)$. Then there exists some x' sufficiently close to x with $x' \succ x$ and $p \cdot x' < w$, which contradicts the fact that $x \in x(p, w)$. Thus $p \cdot x = w$. □



Solving for Marshallian demand I

Suppose the utility function is differentiable

- This is an ungrounded assumption
- However, differentiability can not be falsified by any finite data set
- Also, utility functions are robust to monotone transformations

We may be able to use Kuhn-Tucker to "solve" the UMP:

Utility Maximization Problem

$$\max_{x \in \mathbb{R}^n_+} u(x) \text{ such that } p \cdot x \leq w$$

gives the Lagrangian

$$\mathcal{L}(x, \lambda, \mu, p, w) \equiv u(x) + \lambda(w - p \cdot x) + \mu \cdot x.$$

Solving for Marshallian demand II

1. First order conditions:

$$u_i'(x^*) = \lambda p_i - \mu_i \text{ for all } i$$

2. Complementary slackness:

$$\lambda(w - p \cdot x^*) = 0$$
$$\mu_i x_i^* = 0 \text{ for all } i$$

3. Non-negativity:

$$\lambda \geq 0 \text{ and } \mu_i \geq 0 \text{ for all } i$$

4. Original constraints $p \cdot x^* \leq w$ and $x_i^* \geq 0$ for all i

We can solve this system of equations for certain functional forms of $u(\cdot)$

The power (and limitations) of Kuhn-Tucker

Kuhn-Tucker provides conditions on (x, λ, μ) given (p, w):

1. First order conditions
2. Complementary slackness
3. Non-negativity
4. (Original constraints)

Kuhn-Tucker tells us that if x^* is a solution to the UMP, there exist some (λ, μ) such that these conditions hold

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1. First order conditions
2. Complementary slackness
3. Non-negativity
4. (Original constraints)

Kuhn-Tucker tells us that if x^* is a solution to the UMP, there exist some (λ, μ) such that these conditions hold; however:

- These are only necessary conditions; there may be (x, λ, μ) that satisfy Kuhn-Tucker conditions but do not solve UMP
- If $u(\cdot)$ is concave, conditions are necessary and sufficient

When are Kuhn-Tucker conditions sufficient?

Kuhn-Tucker conditions are necessary and sufficient for a solution (assuming differentiability) as long as we have a "convex problem":

1. The constraint set is convex
   - If each constraint gives a convex set, the intersection is a convex set
   - The set $\{x : g_k(x, \theta) \geq 0\}$ is convex as long as $g_k(\cdot, \theta)$ is a quasiconcave function of x
2. The objective function is concave
   - If we only know the objective is quasiconcave, there are other conditions that ensure Kuhn-Tucker is sufficient

Intuition from Kuhn-Tucker conditions I

Recall (evaluating at the optimum, and for all i):

$$\begin{aligned} \text{FOC} \quad & u_i'(x) = \lambda p_i - \mu_i \\ \text{CS} \quad & \lambda(w - p \cdot x) = 0 \text{ and } \mu_i x_i = 0 \\ \text{NN} \quad & \lambda \geq 0 \text{ and } \mu_i \geq 0 \\ \text{Orig} \quad & p \cdot x \leq w \text{ and } x_i \geq 0 \end{aligned}$$

We can summarize as

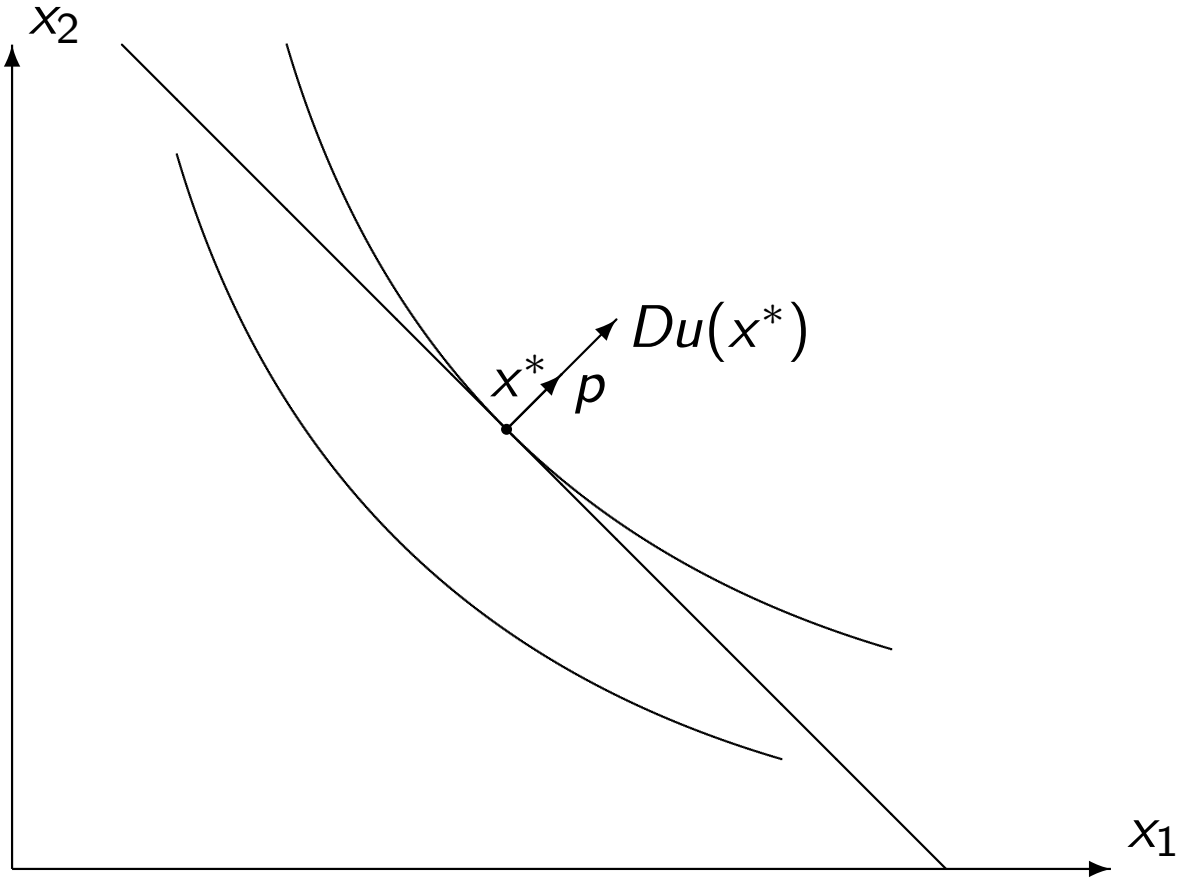
$$u_i'(x) \leq \lambda p_i \text{ with equality if } x_i > 0$$

And therefore if $x_j > 0$ and $x_k > 0$,

$$\frac{p_j}{p_k} = \frac{\frac{\partial u}{\partial x_j}}{\frac{\partial u}{\partial x_k}} \equiv \text{MRS}_{jk}$$

Intuition from Kuhn-Tucker conditions II

- The MRS is the (negative) slope of the indifference curve
- Price ratio is the (negative) slope of the budget line



Intuition from Kuhn-Tucker conditions III

Recall the Envelope Theorem tells us the derivative of the value function in a parameter is the derivative of the Lagrangian:

- Value function (indirect utility)

$$v(p, w) \equiv \sup_{x \in B(p,w)} u(x)$$

- Lagrangian

$$\mathcal{L} \equiv u(x) + \lambda(w - p \cdot x) + \mu \cdot x$$

By the Envelope Theorem, $\frac{\partial v}{\partial w} = \lambda$; i.e., the Lagrange multiplier λ is the "shadow value of wealth" measured in terms of utility

Intuition from Kuhn-Tucker conditions IV

Given our envelope result, we can interpret our earlier condition

$$\frac{\partial u}{\partial x_i} = \lambda p_i \text{ if } x_i > 0$$

as

$$\frac{\partial u}{\partial x_i} = \frac{\partial v}{\partial w} p_i \text{ if } x_i > 0$$

where each side gives the marginal utility from an extra unit of x_i

- LHS directly
- RHS through the wealth we could get by selling it

MRS and separable utility

Recall that if $x_j > 0$ and $x_k > 0$,

$$\text{MRS}_{jk} \equiv \frac{\frac{\partial u}{\partial x_j}}{\frac{\partial u}{\partial x_k}}$$

does not depend on λ; however it typically depends on $x_1, \ldots, x_n$

MRS and separable utility

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does not depend on λ; however it typically depends on $x_1, \ldots, x_n$
Suppose choice from $X \times Y$ where preferences over X do not depend on y

- Recall that $u(x, y) = U\big(v(x), y\big)$ for some $U(\cdot, \cdot)$ and $v(\cdot)$
- $\frac{\partial u}{\partial x_j} = U_1'\big(v(x), y\big)\frac{\partial v}{\partial x_j}$ and $\frac{\partial u}{\partial x_k} = U_1'\big(v(x), y\big)\frac{\partial v}{\partial x_k}$
- $\text{MRS}_{jk} = \frac{\partial v}{\partial x_j} / \frac{\partial v}{\partial x_k}$ does not depend on y

Separability allows empirical work without worrying about y

Recovering Marshallian demand from indirect utility I

To recover the choice correspondence from the value function we typically apply an Envelope Theorem (e.g., Hotelling, Shephard)

- Value function (indirect utility): $v(p, w) \equiv \sup_{x \in B(p,w)} u(x)$
- Lagrangian: $\mathcal{L} \equiv u(x) + \lambda(w - p \cdot x) + \mu \cdot x$

By the ET

$$\frac{\partial v}{\partial w} = \frac{\partial \mathcal{L}}{\partial w} = \lambda$$
$$\frac{\partial v}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} = -\lambda x_i$$

We can combine these, dividing the second by the first...

Recovering Marshallian demand from indirect utility II

Roy's identity

$$x_i(p, w) = -\frac{\frac{\partial v(p,w)}{\partial p_i}}{\frac{\partial v(p,w)}{\partial w}}.$$

We can think of this a little bit like "$\frac{\partial v}{\partial w} = -\frac{\partial v}{x_i \partial p_i}$"

Here we showed Roy's identity as an application of the ET; the notes give an entirely different proof that relies on the expenditure minimization problem

Outline

- The utility maximization problem
  - Marshallian demand and indirect utility
  - First-order conditions of the UMP
  - Recovering demand from indirect utility
- The expenditure minimization problem
- Wealth and substitution effects
  - The Slutsky equation
  - Comparative statics properties

Why we need another "problem"

We would like to characterize "important" properties of Marshallian demand $x(\cdot, \cdot)$ and indirect utility $v(\cdot, \cdot)$

- Unfortunately, this is harder than doing so for $y(\cdot)$ and $\pi(\cdot)$
- Difficulty arises from the fact that in UMP parameters enter feasible set rather than objective

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We would like to characterize "important" properties of Marshallian demand $x(\cdot, \cdot)$ and indirect utility $v(\cdot, \cdot)$

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- Difficulty arises from the fact that in UMP parameters enter feasible set rather than objective

Consider an price increase for one good (apples)

1. Substitution effect: Apples are now relatively more expensive than bananas, so I buy fewer apples
2. Wealth effect: I feel poorer, so I buy _____ (more? fewer?) apples

Wealth effect and substitution effects could go in opposite directions $\implies$ can't easily sign the change in consumption

Isolating the substitution effect

We can isolate the substitution effect by "compensating" the consumer so that her maximized utility does not change

If maximized utility doesn't change, the consumer can't feel richer or poorer; demand changes can therefore be attributed entirely to the substitution effect

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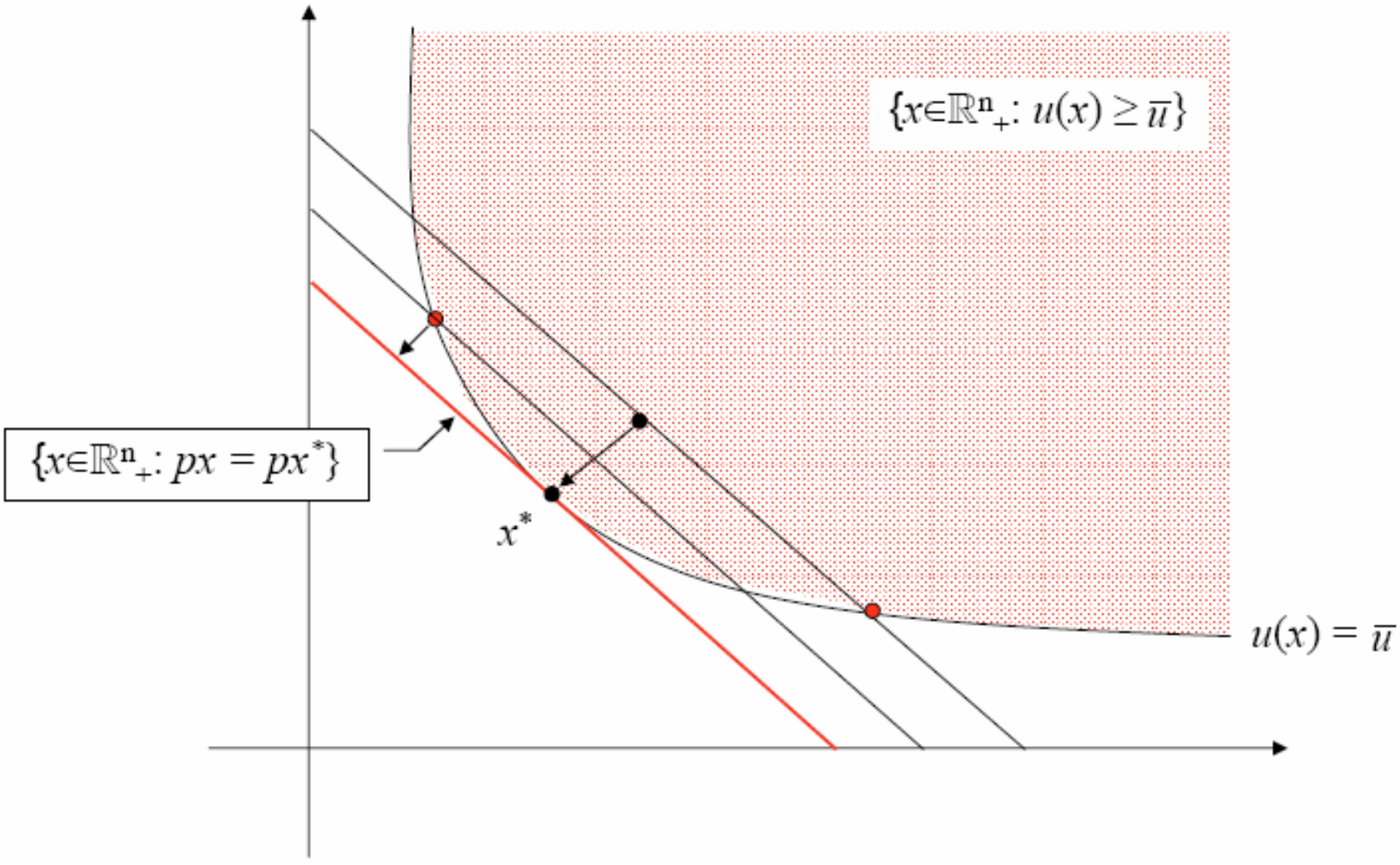
If maximized utility doesn't change, the consumer can't feel richer or poorer; demand changes can therefore be attributed entirely to the substitution effect

Expenditure Minimization Problem

$$\min_{x \in \mathbb{R}^n_+} p \cdot x \text{ such that } u(x) \geq \bar{u}.$$

i.e., find the cheapest bundle at prices p that yield utility at least $\bar{u}$

Illustrating the Expenditure Minimization Problem



Expenditure minimization problem

The consumer's Hicksian demand is given by correspondence $h\colon \mathbb{R}^n \times \mathbb{R} \rightrightarrows \mathbb{R}^n$

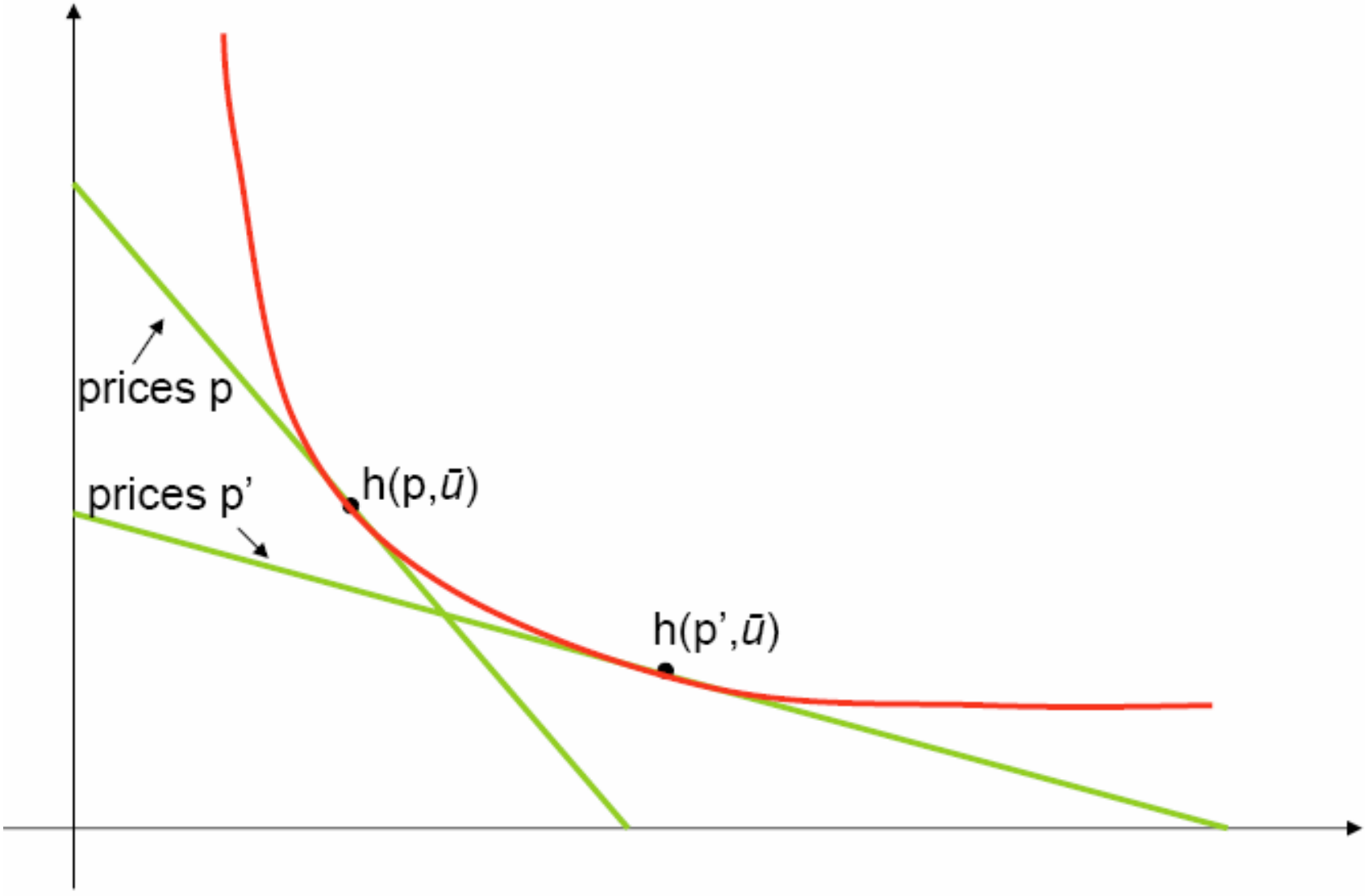
$$\begin{aligned} h(p,\bar{u}) &\equiv \underset{x\in\mathbb{R}^n_+ :\, u(x)\geq\bar{u}}{\operatorname{argmin}} \; p\cdot x \\ &= \{x \in \mathbb{R}^n_+ :\, u(x) \geq \bar{u} \text{ and } p\cdot x = e(p,\bar{u})\} \end{aligned}$$

Resulting expenditure function is given by

$$e(p,\bar{u}) \equiv \min_{x\in\mathbb{R}^n_+ :\, u(x)\geq\bar{u}} p\cdot x$$

Note we have used min instead of inf assuming conditions (listed in the notes) under which a minimum *is* achieved

Illustrating Hicksian demand



Relating Hicksian and Marshallian demand I

Theorem ("Same problem" identities)

Suppose $u(\cdot)$ is a utility function representing a continuous and locally non-satiated preference relation $\succsim$ on $\mathbb{R}^n_+$. Then for any $p \gg \mathbf{0}$ and $w \geq 0$,

1. $h\big(p, v(p,w)\big) = x(p,w)$,
2. $e\big(p, v(p,w)\big) = w$;

and for any $\bar{u} \geq u(\mathbf{0})$,

3. $x\big(p, e(p,\bar{u})\big) = h(p,\bar{u})$, *and*
4. $v\big(p, e(p,\bar{u})\big) = \bar{u}$.

For proofs see notes (cumbersome but relatively straightforward)

Relating Hicksian and Marshallian demand II

These say that UMP and EMP are fundamentally solving the same problem, so:

- If the utility you can get with wealth w is $v(p, w)$...
  - To achieve utility $v(p, w)$ will cost at least w
  - You will buy the same bundle whether you have w to spend, or you are trying to achieve utility $v(p, w)$
- If it costs $e(p, \bar{u})$ to achieve utility $\bar{u}$...
  - Given wealth $e(p, \bar{u})$ you will achieve utility at most $\bar{u}$
  - You will buy the same bundle whether you have $e(p, \bar{u})$ to spend, or you are trying to achieve utility $\bar{u}$

The EMP should look familiar...

Expenditure Minimization Problem

$$\min_{x \in \mathbb{R}^n_+} p \cdot x \text{ such that } u(x) \geq \bar{u}.$$

The EMP should look familiar...

Expenditure Minimization Problem

$$\min_{x \in \mathbb{R}^n_+} p \cdot x \text{ such that } u(x) \geq \bar{u}.$$

Recall

Single-output Cost Minimization Problem

$$\min_{z \in \mathbb{R}^m_+} w \cdot z \text{ such that } f(z) \geq q.$$

If we interpret $u(\cdot)$ as the production function of the consumer's "hedonic firm," these are the same problem

All of our CMP results go through...

Properties of Hicksian demand and expenditure I

As in our discussion of the single-output CMP:

- $e(p, \bar{u}) = p \cdot h(p, \bar{u})$ (adding up)
- $e(\cdot, \bar{u})$ is homogeneous of degree one in p
- $h(\cdot, \bar{u})$ is homogeneous of degree zero in p
- If $e(\cdot, \bar{u})$ is differentiable in p, then $\nabla_p e(p, \bar{u}) = h(p, \bar{u})$ (Shephard's Lemma)
- $e(\cdot, \bar{u})$ is concave in p
- If $h(\cdot, \bar{u})$ is differentiable in p, then the matrix $D_p h(p, \bar{u}) = D_p^2 e(p, \bar{u})$ is symmetric and negative semidefinite
- $e(p, \cdot)$ is nondecreasing in $\bar{u}$
- Rationalizability condition...

Properties of Hicksian demand and expenditure II

Theorem

Hicksian demand function $h\colon P \times \mathbb{R} \rightrightarrows \mathbb{R}^n_+$ *and differentiable expenditure function* $e\colon P \times \mathbb{R} \to \mathbb{R}$ *on an open convex set* $P \subseteq \mathbb{R}^n$ *of prices are jointly rationalizable for a fixed utility* $\bar{u}$ *of a monotone utility function iff*

1. $e(p, \bar{u}) = p \cdot h(p, \bar{u})$ *(adding-up);*
2. $\nabla_p e(p, \bar{u}) = h(p, \bar{u})$ *(Shephard's Lemma);*
3. $e(p, \bar{u})$ *is concave in* p *(for a fixed* $\bar{u}$*).*

The Slutsky Matrix

Definition (Slutsky matrix)

$$D_p h(p, \bar{u}) \equiv \left[\frac{\partial h_i(p, \bar{u})}{\partial p_j} \right]_{i,j} \equiv \begin{bmatrix} \frac{\partial h_1(p,\bar{u})}{\partial p_1} & \cdots & \frac{\partial h_1(p,\bar{u})}{\partial p_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n(p,\bar{u})}{\partial p_1} & \cdots & \frac{\partial h_n(p,\bar{u})}{\partial p_n} \end{bmatrix}.$$

- Concavity of $e(\cdot, \bar{u})$ and Shephard's Lemma give that the Slutsky matrix is symmetric and negative semidefinite (as we found for the substitution matrix)
- $h(\cdot, \bar{u})$ is homogeneous of degree zero in p, so by Euler's Law

$$D_p h(p, \bar{u})\, p = \mathbf{0}$$

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- The utility maximization problem
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Relating (changes in) Hicksian and Marshallian demand

Assuming differentiability and hence single-valuedness, we can differentiate the ith row of the identity

$$h(p, \bar{u}) = x\big(p, e(p, \bar{u})\big)$$

in p_j to get

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial w} \underbrace{\frac{\partial e}{\partial p_j}}_{=h_j=x_j}$$

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial w} x_j$$

The Slutsky equation I

Slutsky equation

$$\underbrace{\frac{\partial x_i(p,w)}{\partial p_j}}_{\text{total effect}} = \underbrace{\frac{\partial h_i\big(p,u(x(p,w))\big)}{\partial p_j}}_{\text{substitution effect}} - \underbrace{\frac{\partial x_i(p,w)}{\partial w}x_j(p,w)}_{\text{wealth effect}}$$

for all i and j.

In matrix form, we can instead write

$$\nabla_p x = \nabla_p h - (\nabla_w x)x^\top.$$

The Slutsky equation II

Setting $i = j$, we can decompose the effect of an an increase in p_i

$$\frac{\partial x_i(p,w)}{\partial p_i} = \frac{\partial h_i\big(p, u(x(p,w))\big)}{\partial p_i} - \frac{\partial x_i(p,w)}{\partial w} x_i(p,w)$$

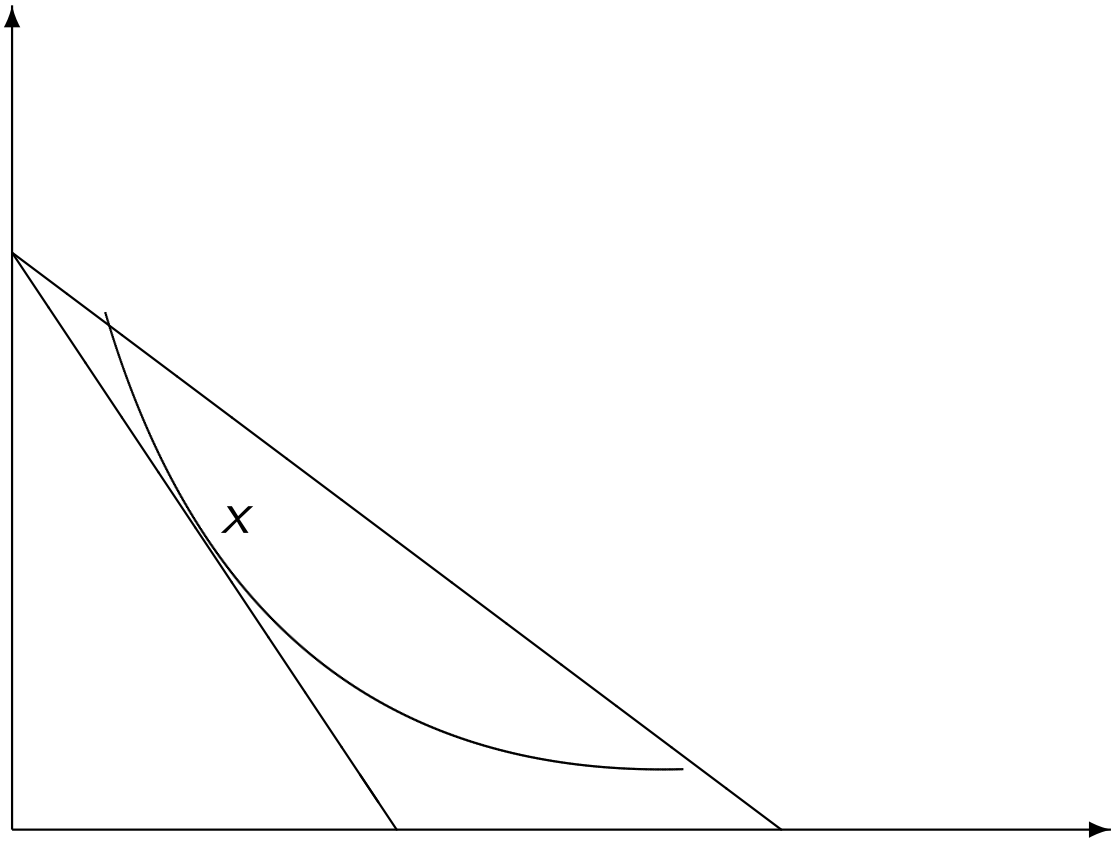
An "own-price" increase...

1. Encourages consumer to substitute away from good i
   - $\frac{\partial h_i}{\partial p_i} \leq 0$ by negative semidefiniteness of Slutsky matrix
2. Makes consumer poorer, which affects consumption of good i in some indeterminate way
   - Sign of $\frac{\partial x_i}{\partial w}$ depends on preferences

Illustrating wealth and substitution effects

Following a decrease in the price of the first good...

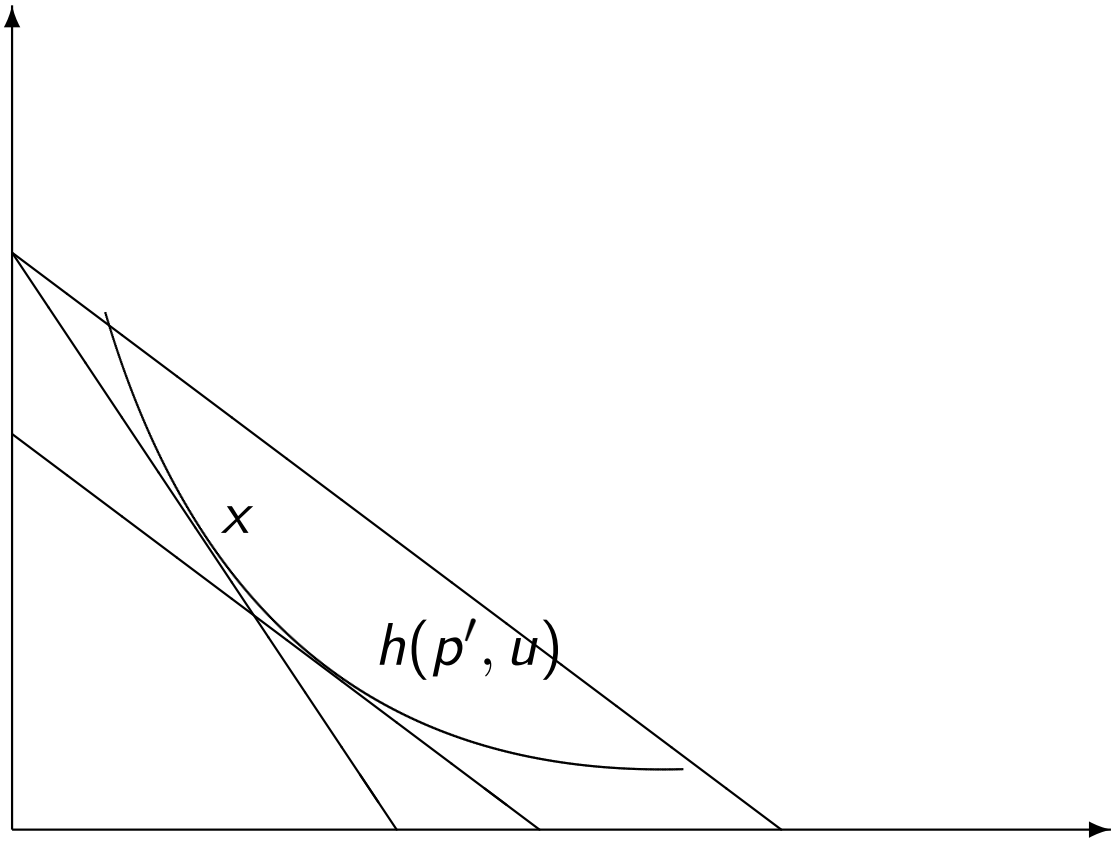
- Substitution effect moves from x to h
- Wealth effect moves from h to x'



Illustrating wealth and substitution effects

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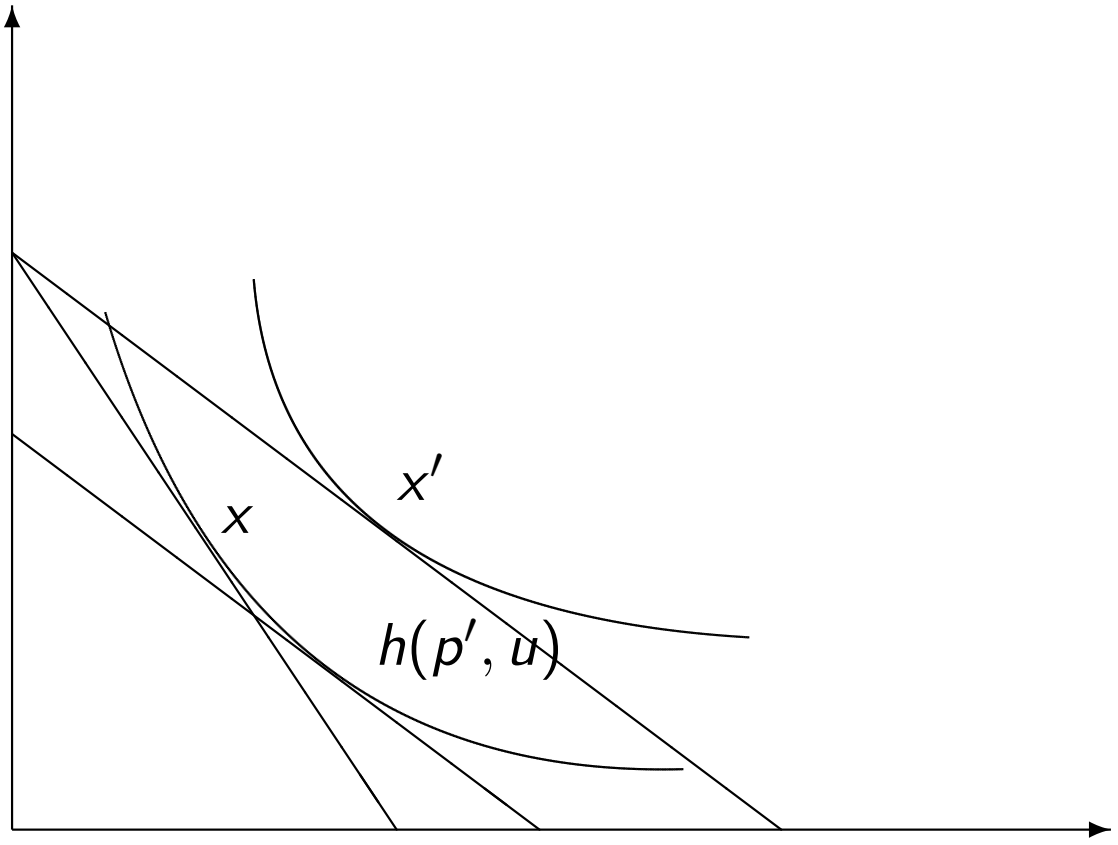
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Illustrating wealth and substitution effects

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Marshallian response to changes in wealth

Definition (Normal good)

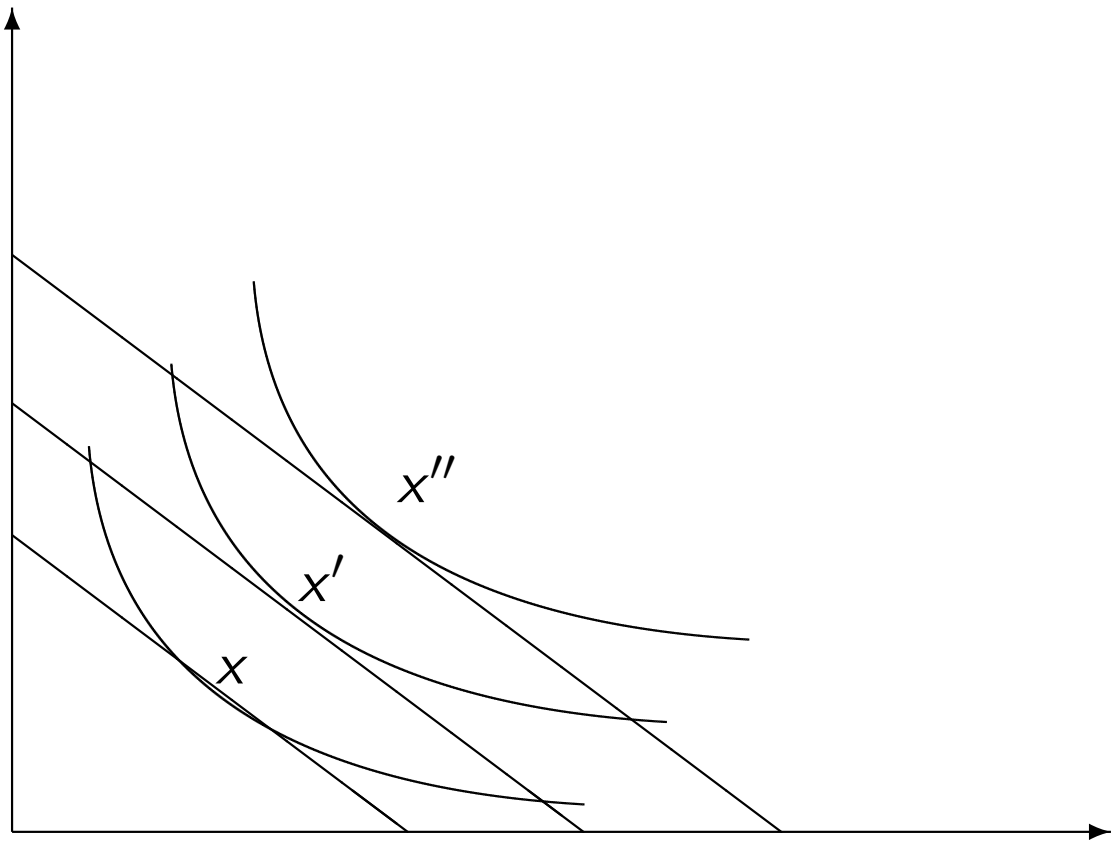
Good i is a normal good if $x_i(p, w)$ is increasing in w.

Definition (Inferior good)

Good i is an inferior good if $x_i(p, w)$ is decreasing in w.

Graphing Marshallian response to changes in wealth

- Engle curves show how Marshallian demand moves with wealth (locus of $\{x, x', x'', \dots\}$ below)
- In this example, both goods are normal (x_i increases in w)



Marshallian response to changes in own price

Definition (Regular good)

Good i is a regular good if $x_i(p, w)$ is decreasing in p_i.

Definition (Giffen good)

Good i is a Giffen good if $x_i(p, w)$ is increasing in p_i.

Potatoes during the Irish potato famine are the canonical example (and probably weren't *actually* Giffen goods)

Marshallian response to changes in own price

Definition (Regular good)

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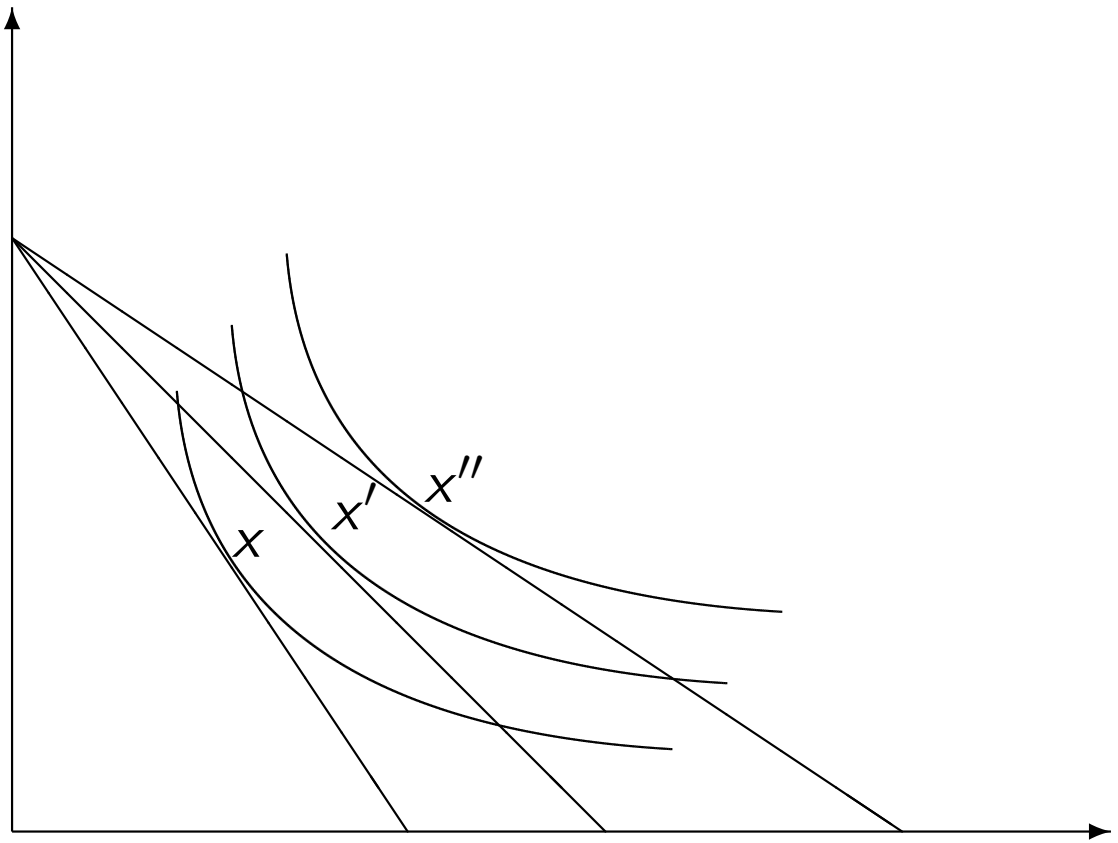
Potatoes during the Irish potato famine are the canonical example (and probably weren't *actually* Giffen goods)

By the Slutsky equation (which gives $\frac{\partial x_i}{\partial p_i} = \frac{\partial h_i}{\partial p_i} - \frac{\partial x_i}{\partial w} x_i$ for $i = j$)

- Normal $\implies$ regular
- Giffen $\implies$ inferior

Graphing Marshallian response to changes in own price

- Offer curves show how Marshallian demand moves with price
- In this example, good 1 is regular and good 2 is a gross complement for good 1



Marshallian response to changes in other goods' price

Definition (Gross substitute)

Good i is a gross substitute for good j if $x_i(p, w)$ is increasing in p_j.

Definition (Gross complement)

Good i is a gross complement for good j if $x_i(p, w)$ is decreasing in p_j.

Gross substitutability/complementarity is not necessarily symmetric

Hicksian response to changes in other goods' price

Definition (Substitute)

Good i is a substitute for good j if $h_i(p, \bar{u})$ is increasing in p_j.

Definition (Complement)

Good i is a complement for good j if $h_i(p, \bar{u})$ is decreasing in p_j.

Substitutability/complementarity is symmetric

In a two-good world, the goods must be substitutes (*why?*)

Part VI

Consumer Theory 2

Recap: The consumer problems

Utility Maximization Problem

$$\max_{x \in \mathbb{R}^n_+} u(x) \text{ such that } p \cdot x \leq w.$$

- Choice correspondence: Marshallian demand $x(p, w)$
- Value function: indirect utility function $v(p, w)$

Expenditure Minimization Problem

$$\min_{x \in \mathbb{R}^n_+} p \cdot x \text{ such that } u(x) \geq \bar{u}.$$

- Choice correspondence: Hicksian demand $h(p, \bar{u})$
- Value function: expenditure function $e(p, \bar{u})$

Key questions addressed by consumer theory

Already addressed

- What problems do consumers solve?
- What do we know about the solutions to these CPs generally? What about if we apply restrictions to preferences?
- How do we actually solve these CPs?
- How do the value functions and choice correspondences relate within/across UMP and EMP?

Key questions addressed by consumer theory

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- What problems do consumers solve?
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Still to come

- How do we measure consumer welfare?
- How should we calculate price indices?
- When and how can we aggregate across heterogeneous consumers?

Outline

- The welfare impact of price changes
- Price indices
  - Price indices for all goods
  - Price indices for a subset of goods
- Aggregating across consumers
- Optimal taxation

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Quantifying consumer welfare I

Key question

How much better or worse off is a consumer as a result of a price change from p to p'?

Applies broadly:

- Actual price changes
- Taxes or subsidies
- Introduction of new goods

Quantifying consumer welfare II

Challenge will be to measure how "well off" a consumer is without using utils—recall preference representation is ordinal
This rules out a first attempt:

$$\Delta u = v(p', w) - v(p, w)$$

To get a dollar-denominated measure, we can ask one of two questions:

1. How much would consumer be willing to pay for the price change?
   Fee + Price change $\sim$ Status quo
2. How much would we have to pay consumer to miss out on price change?
   Price change $\sim$ Status quo + Bonus

Quantifying consumer welfare III

Both questions fundamentally ask "how much money is required to achieve a fixed level of utility before and after the price change?"

$$\text{Variation} = e(p, u_{\text{reference}}) - e(p', u_{\text{reference}})$$

For our two questions,

1. How much would consumer be willing to pay for the price change?
   Reference: Old utility ($u_{\text{reference}} = \bar{u} \equiv v(p, w)$)
2. How much would we have to pay consumer to miss out on price change?
   Reference: New utility ($u_{\text{reference}} = \bar{u}' \equiv v(p', w)$)

Compensating and equivalent variation

Definition (Compensating variation)

The amount less wealth (i.e., the fee) a consumer needs to achieve the same maximum utility at new prices (p') as she had before the price change (at prices p):

$$\text{CV} \equiv e\big(p, v(p,w)\big) - e\big(p', v(p,w)\big) = w - e\big(p', \underbrace{v(p,w)}_{\equiv \bar{u}}\big).$$

Compensating and equivalent variation

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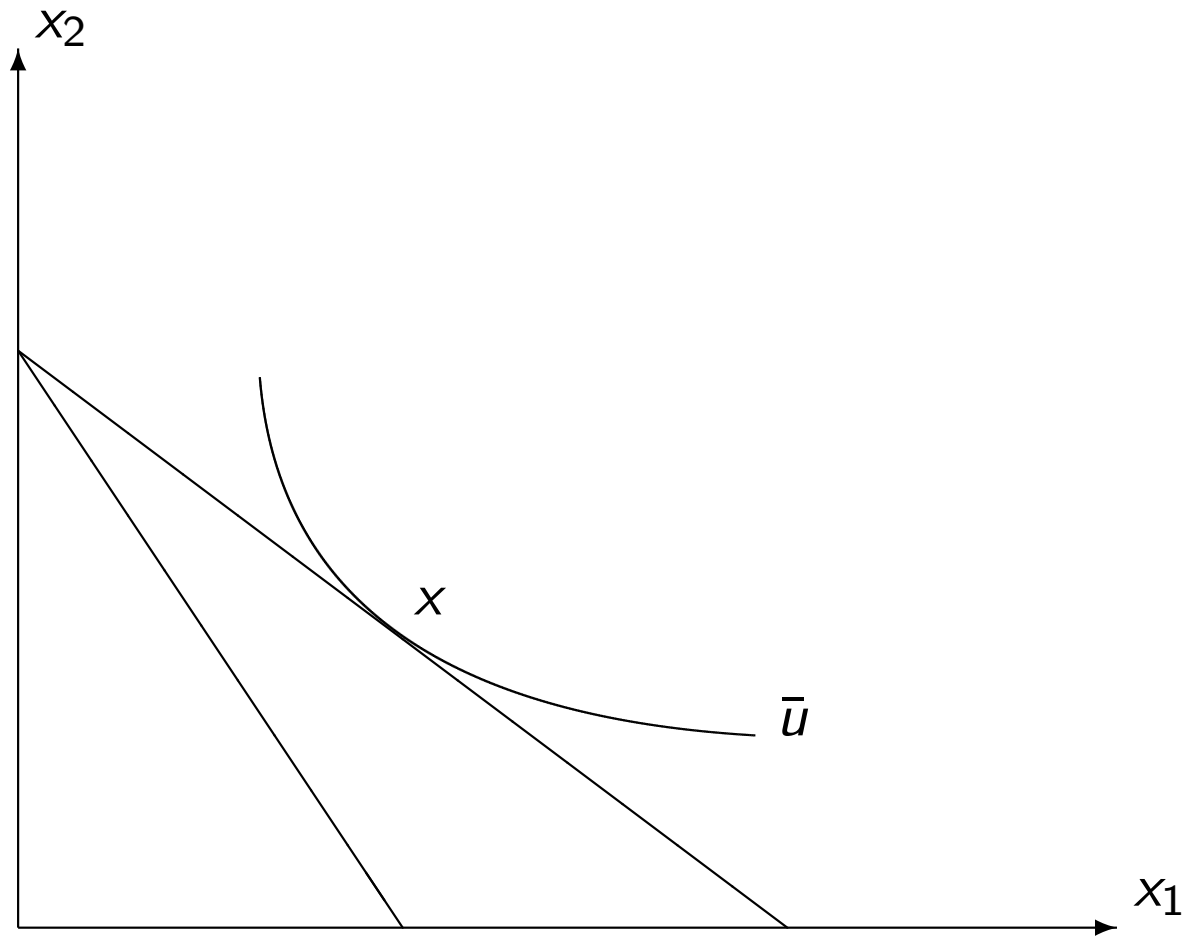
Definition (Equivalent variation)

The amount more wealth (i.e., the bonus) a consumer needs to achieve the same maximum utility at old prices (p) as she could achieve after a price change (to p'):

$$\text{EV} \equiv e\big(p, v(p',w)\big) - e\big(p', v(p',w)\big) = e\big(p, \underbrace{v(p',w)}_{\equiv \bar{u}'}\big) - w.$$

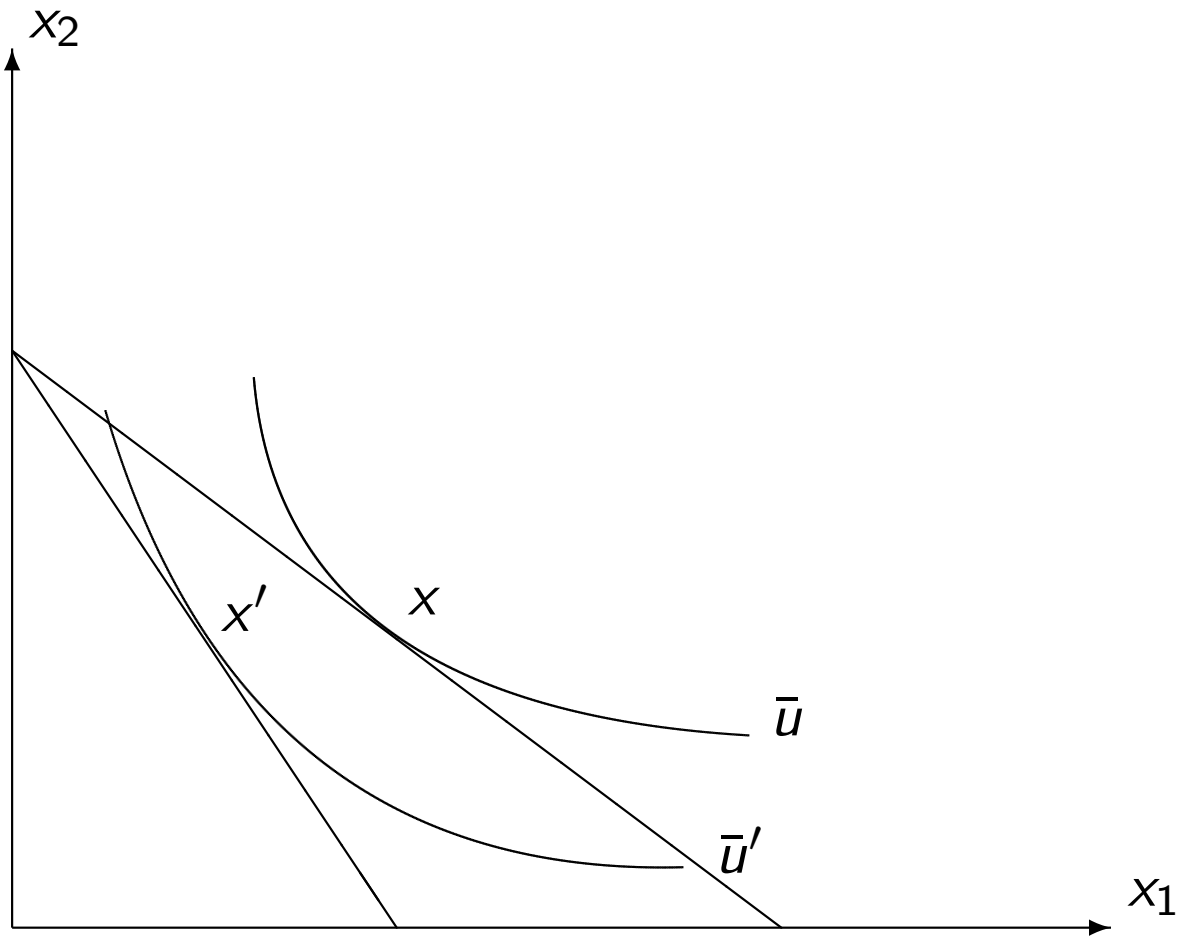
Illustrating compensating variation

- Suppose the price of good two is 1
- Price of good one increases



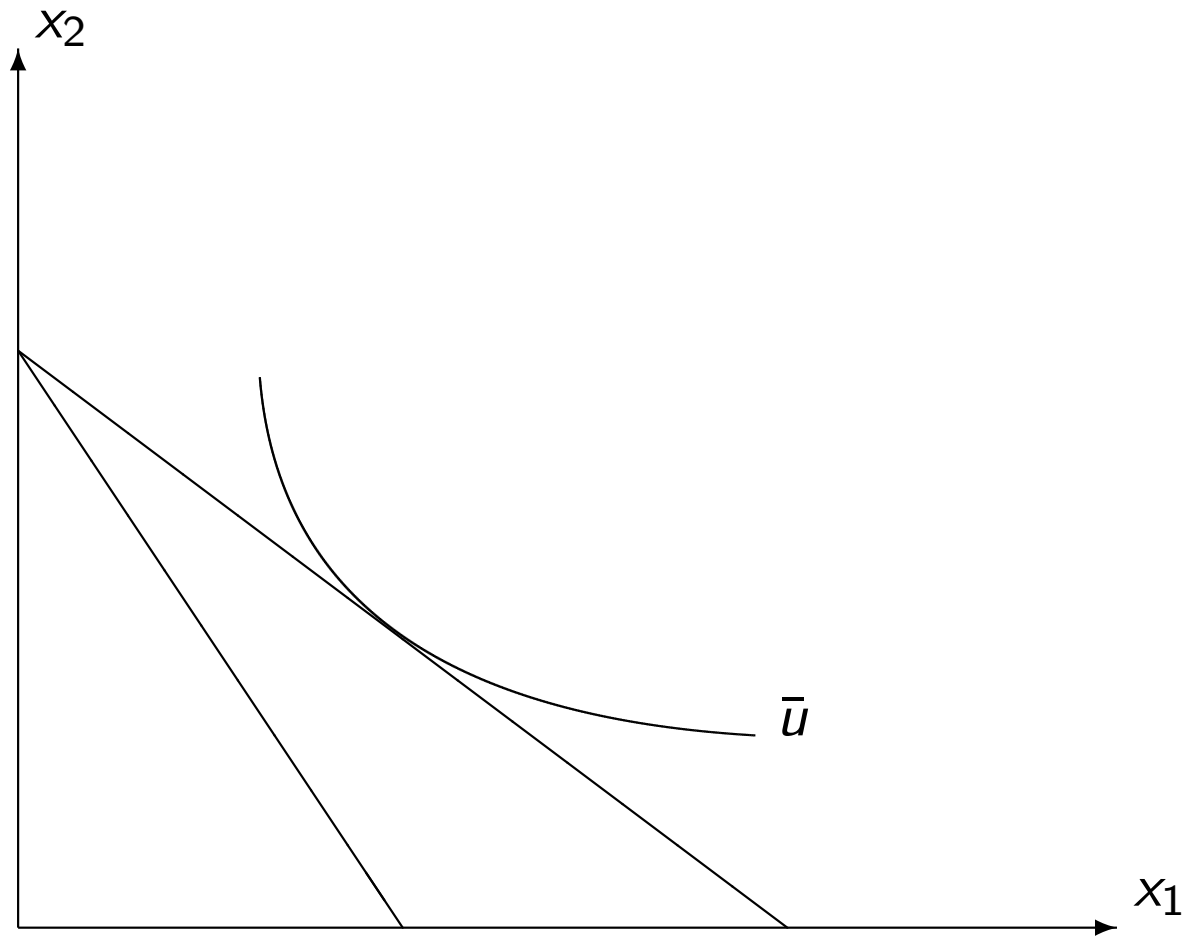
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- Price of good one increases



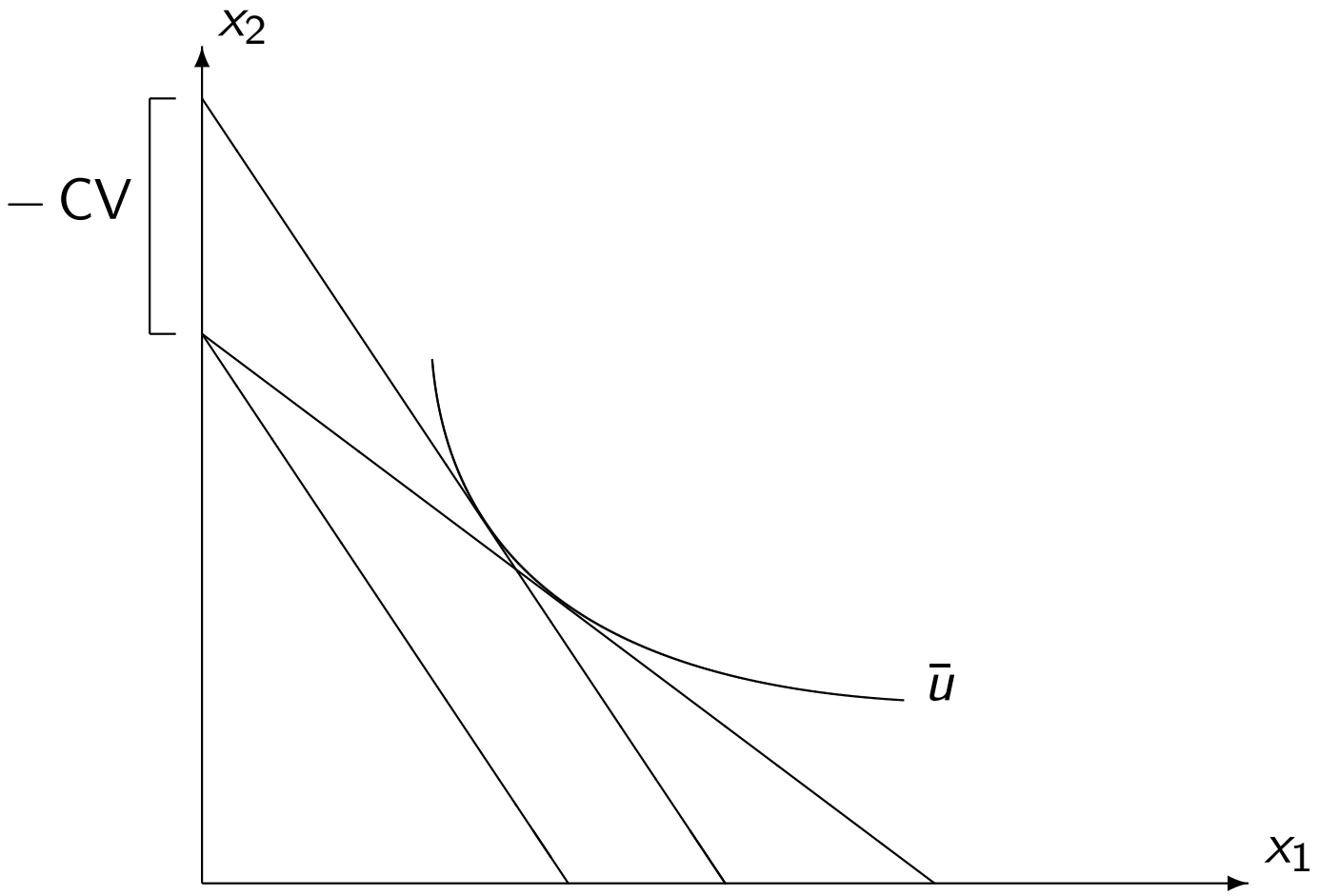
Illustrating compensating variation

- Suppose the price of good two is 1
- Price of good one increases



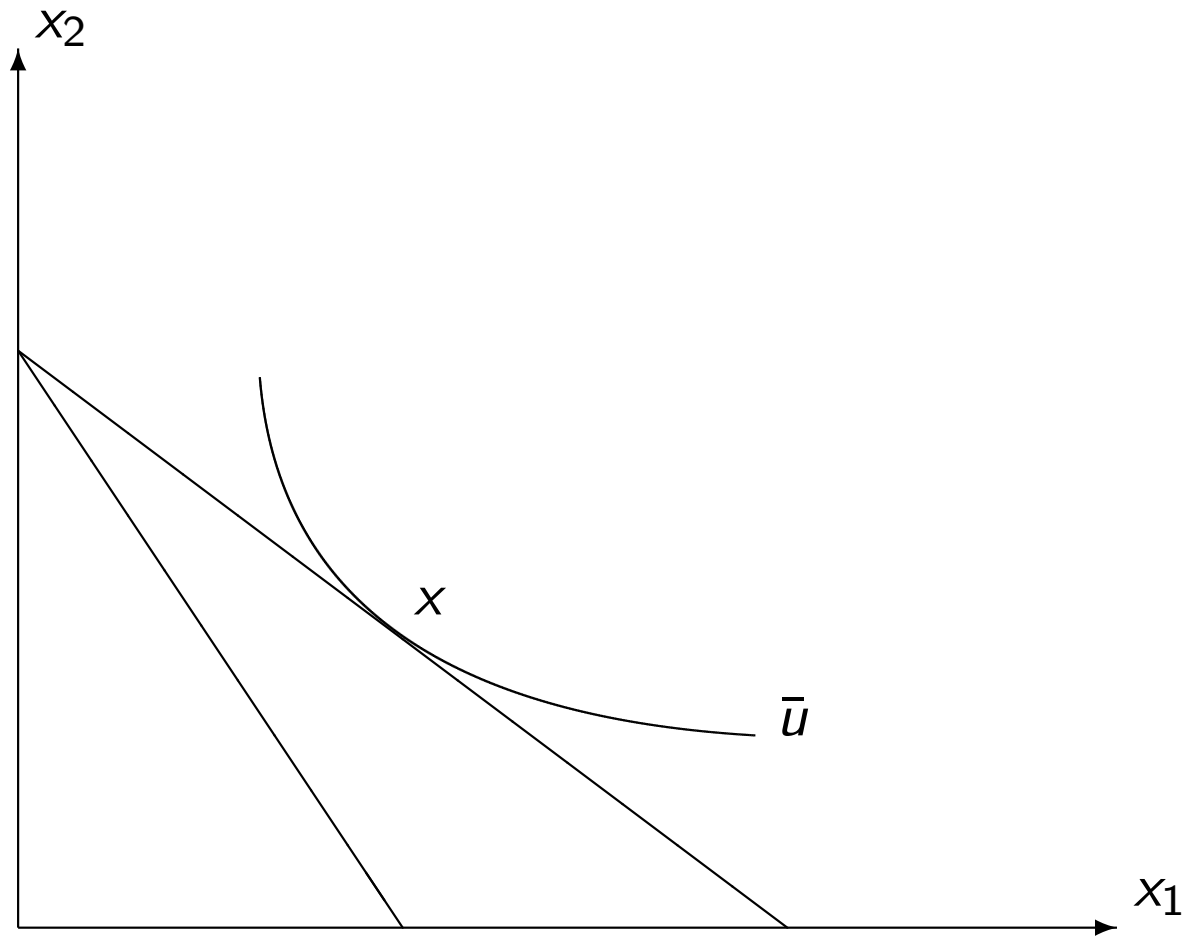
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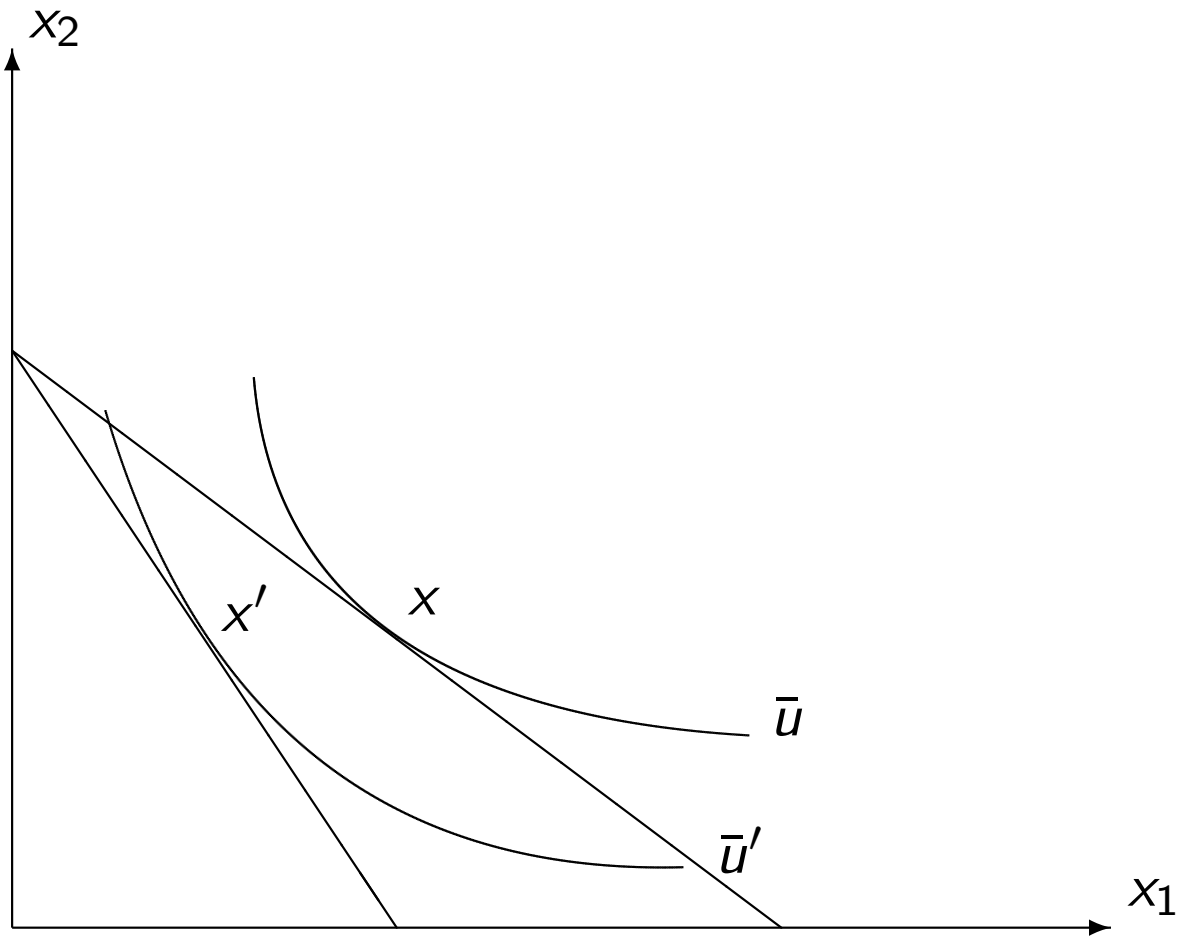
Illustrating equivalent variation

- Suppose the price of good two is 1
- Price of good one increases



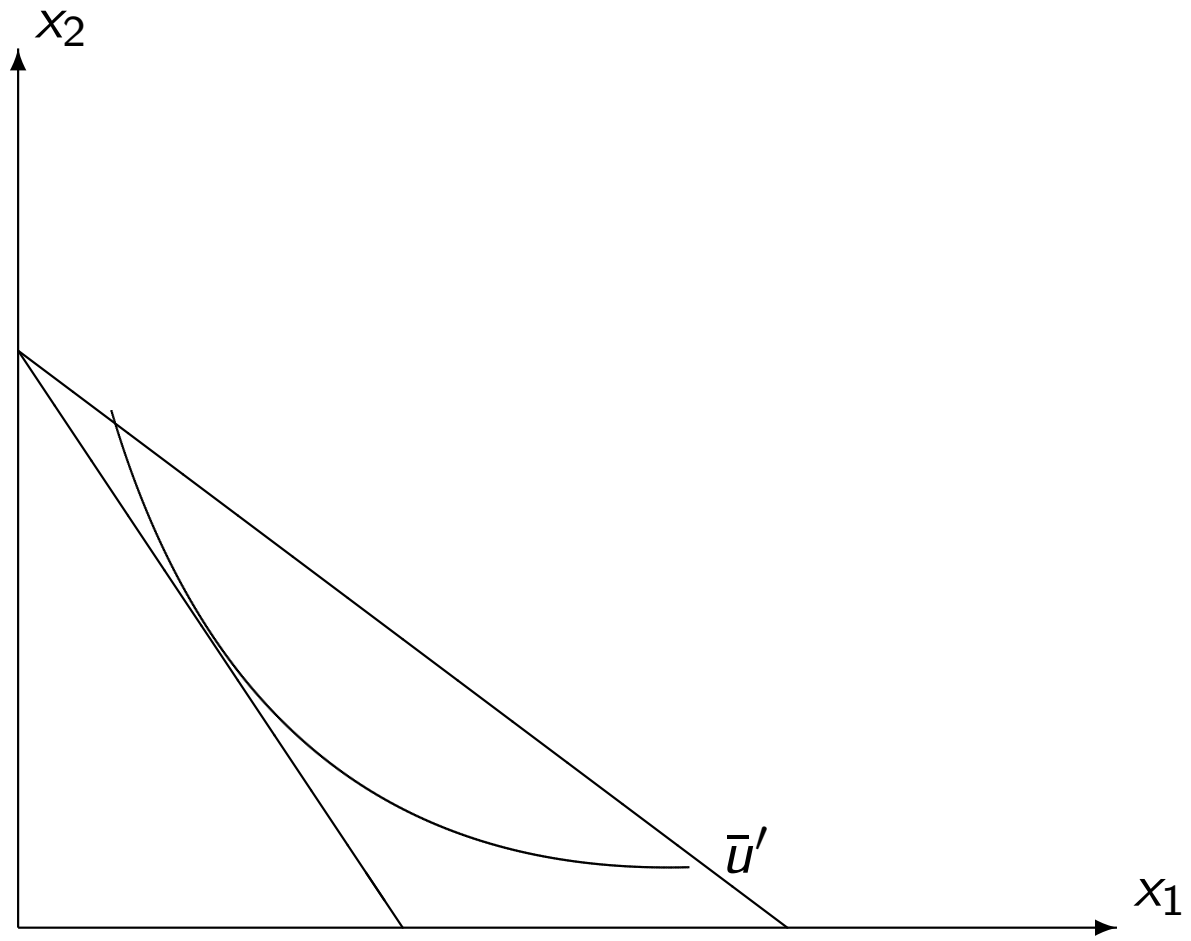
Illustrating equivalent variation

- Suppose the price of good two is 1
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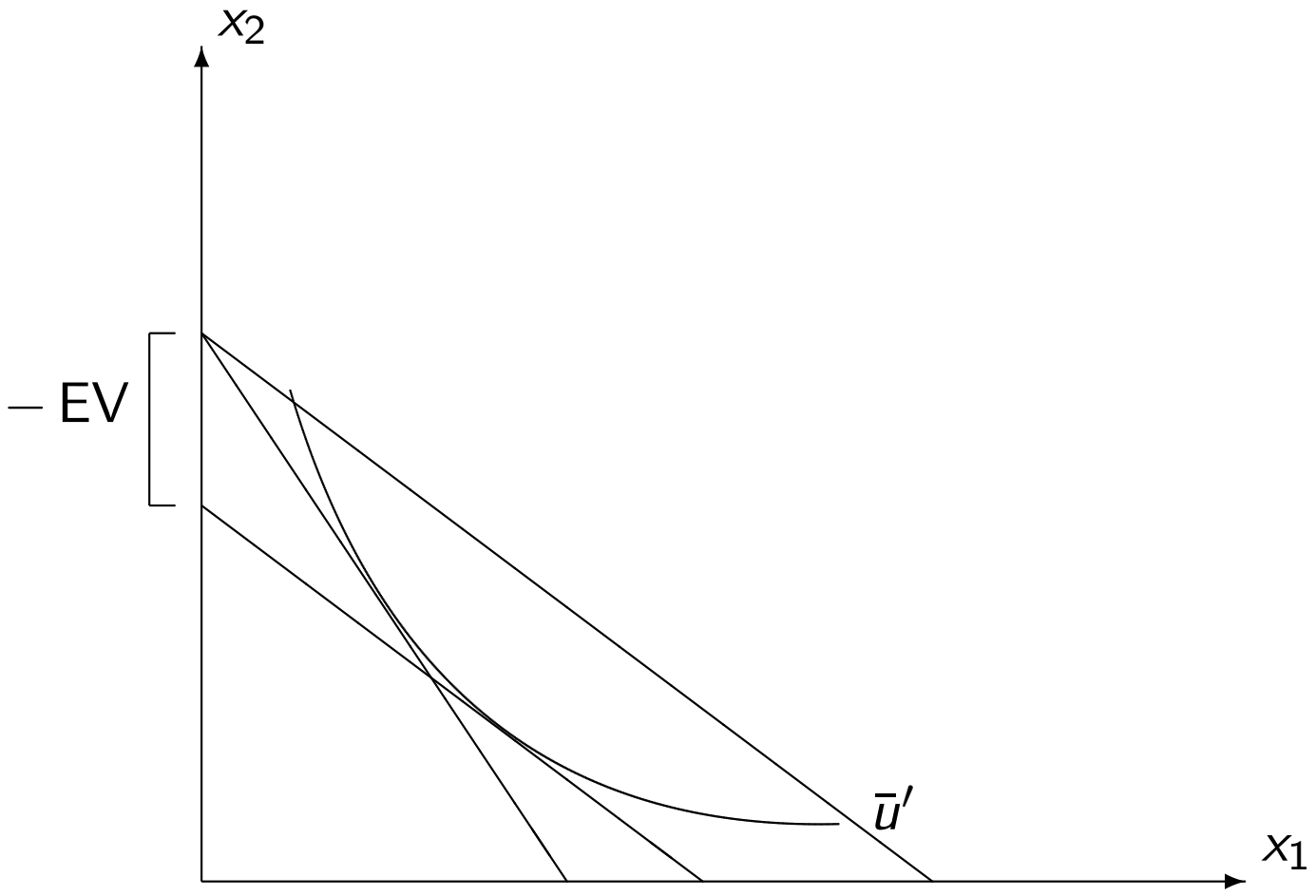
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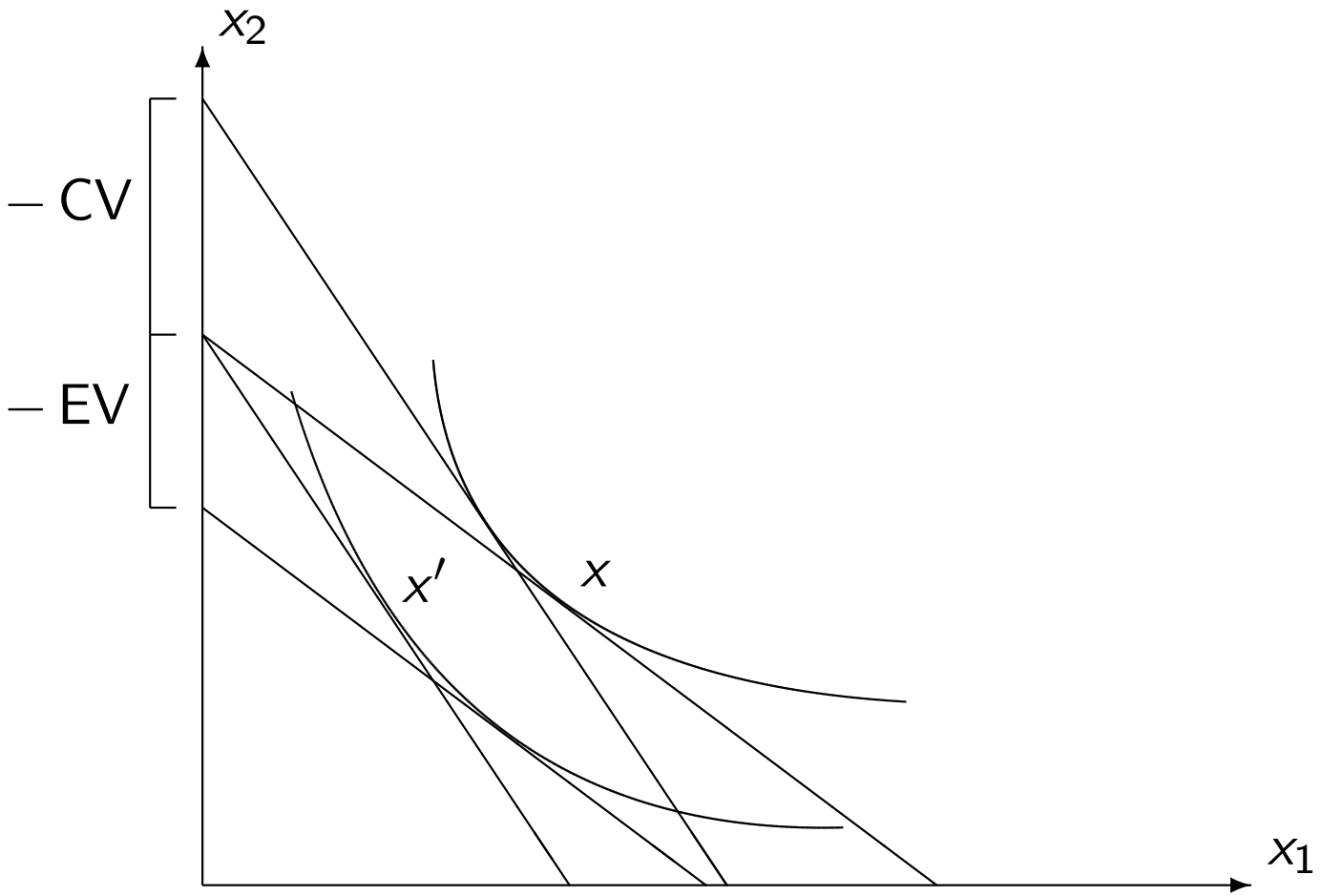
Illustrating equivalent variation

- Suppose the price of good two is 1
- Price of good one increases



We can't order CV and EV

- CV and EV are not necessarily equal
- We can't generally say which is bigger



Changing prices for a single good

Recall

$$\text{CV} = e(p, \bar{u}) - e(p', \bar{u})$$

Suppose the price of a single good changes from $p_i \longrightarrow p_i'$

$$= \int_{p_i'}^{p_i} \frac{\partial e(p, \bar{u})}{\partial p_i} \, dp_i$$

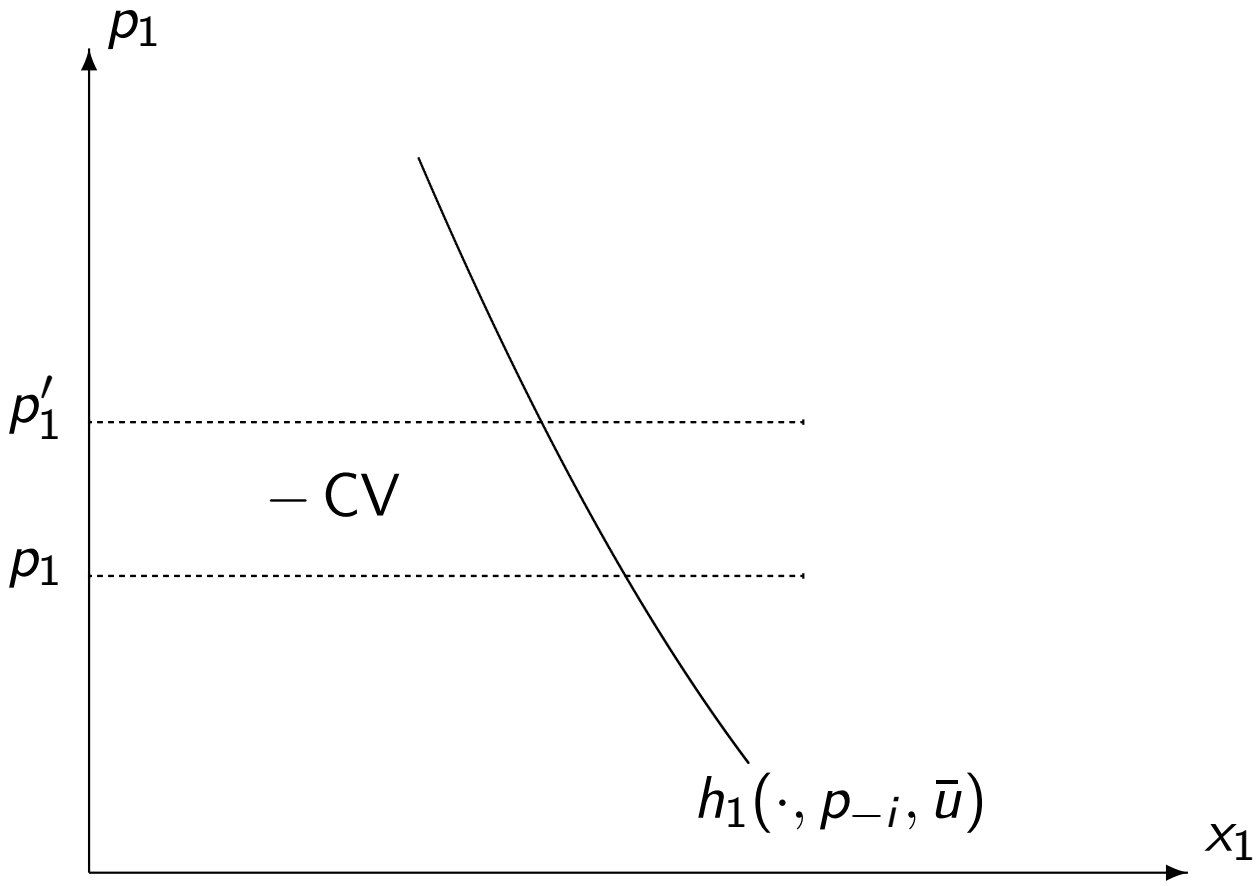
$$= \int_{p_i'}^{p_i} h_i(p, \bar{u}) \, dp_i = - \int_{p_i}^{p_i'} h_i(p, \bar{u}) \, dp_i$$

Similarly,

$$\text{EV} = \int_{p_i'}^{p_i} h_i(p, \bar{u}') \, dp_i = - \int_{p_i}^{p_i'} h_i(p, \bar{u}') \, dp_i$$

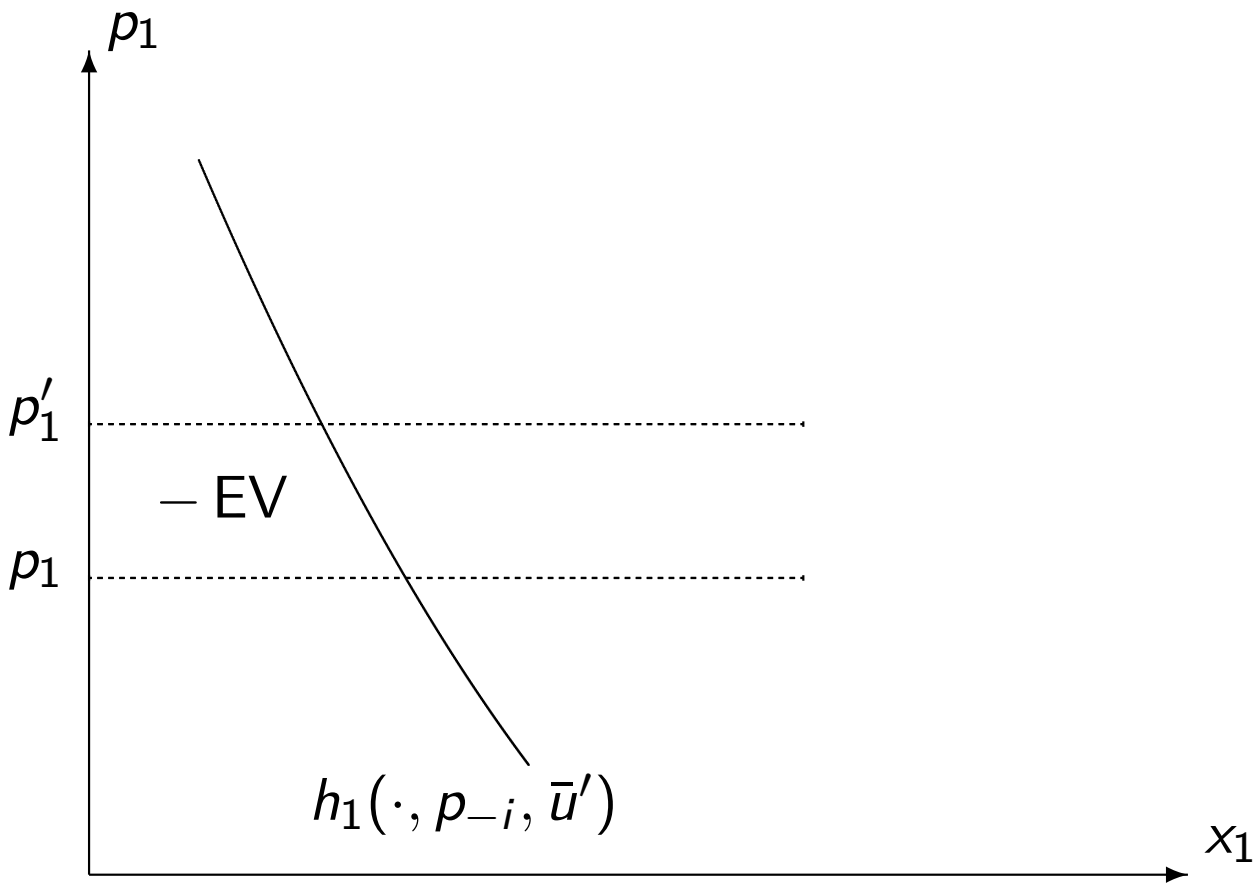
Illustrating changing prices for a single good: CV

- Suppose the price of good one increases from p_1 to p_1'
- Let $\bar{u} \equiv v(p, w)$ and $\bar{u}' \equiv v(p', w)$



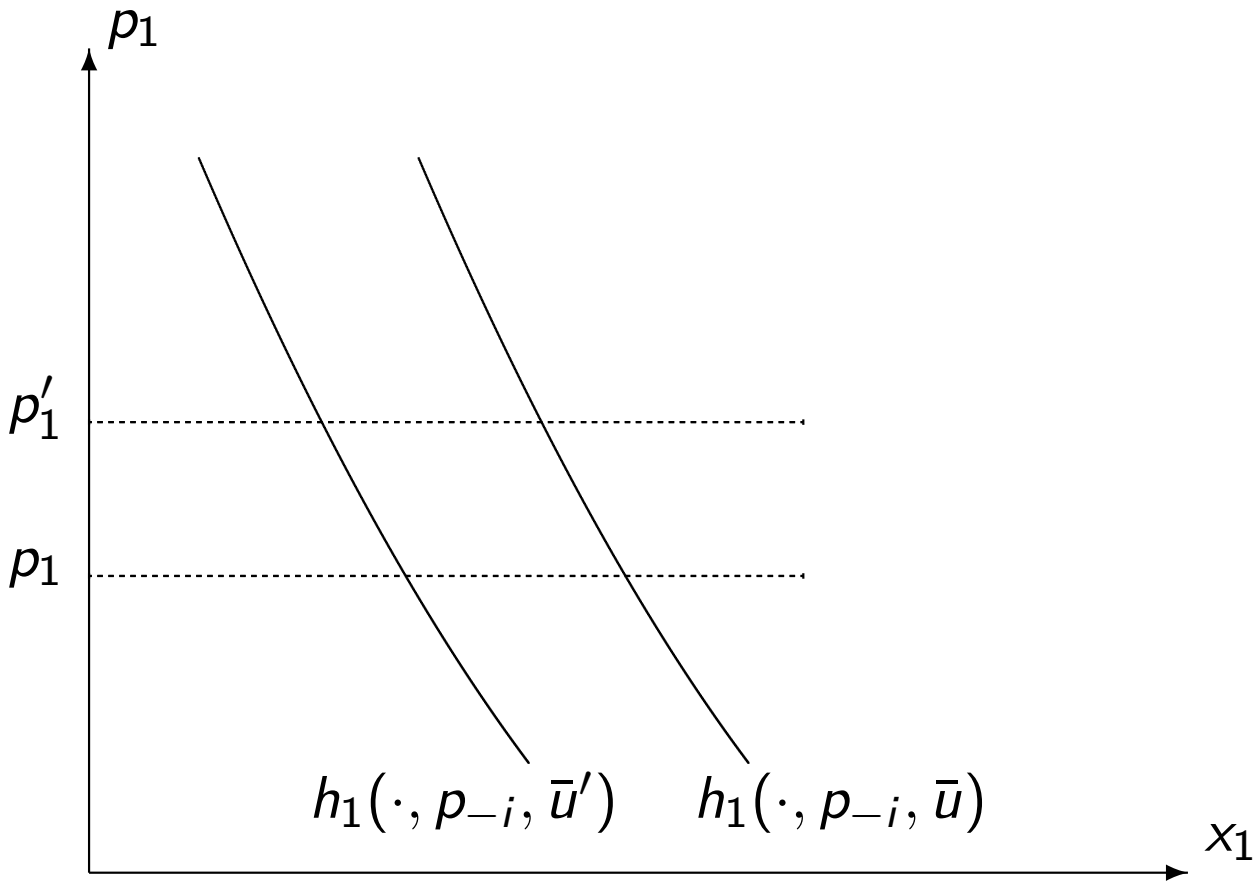
Illustrating changing prices for a single good: EV

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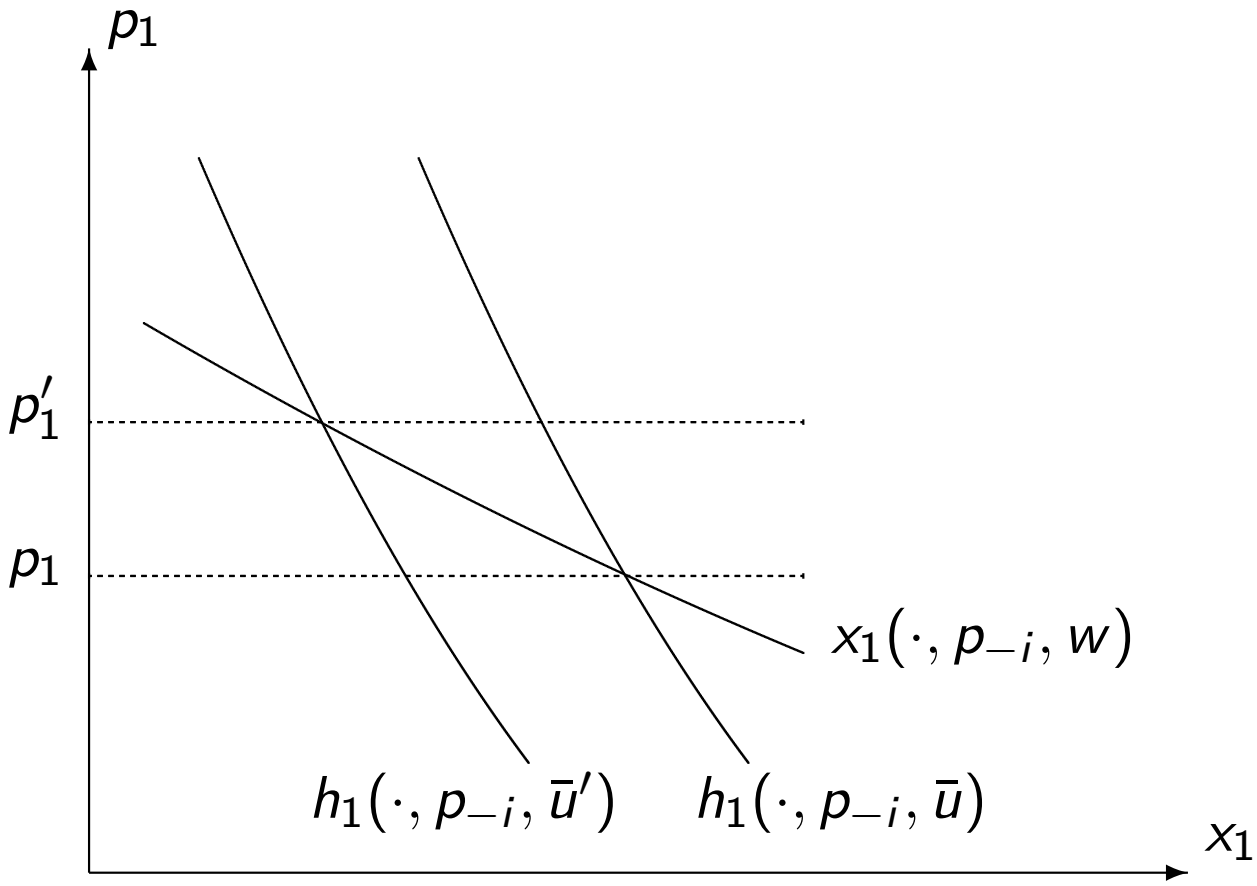
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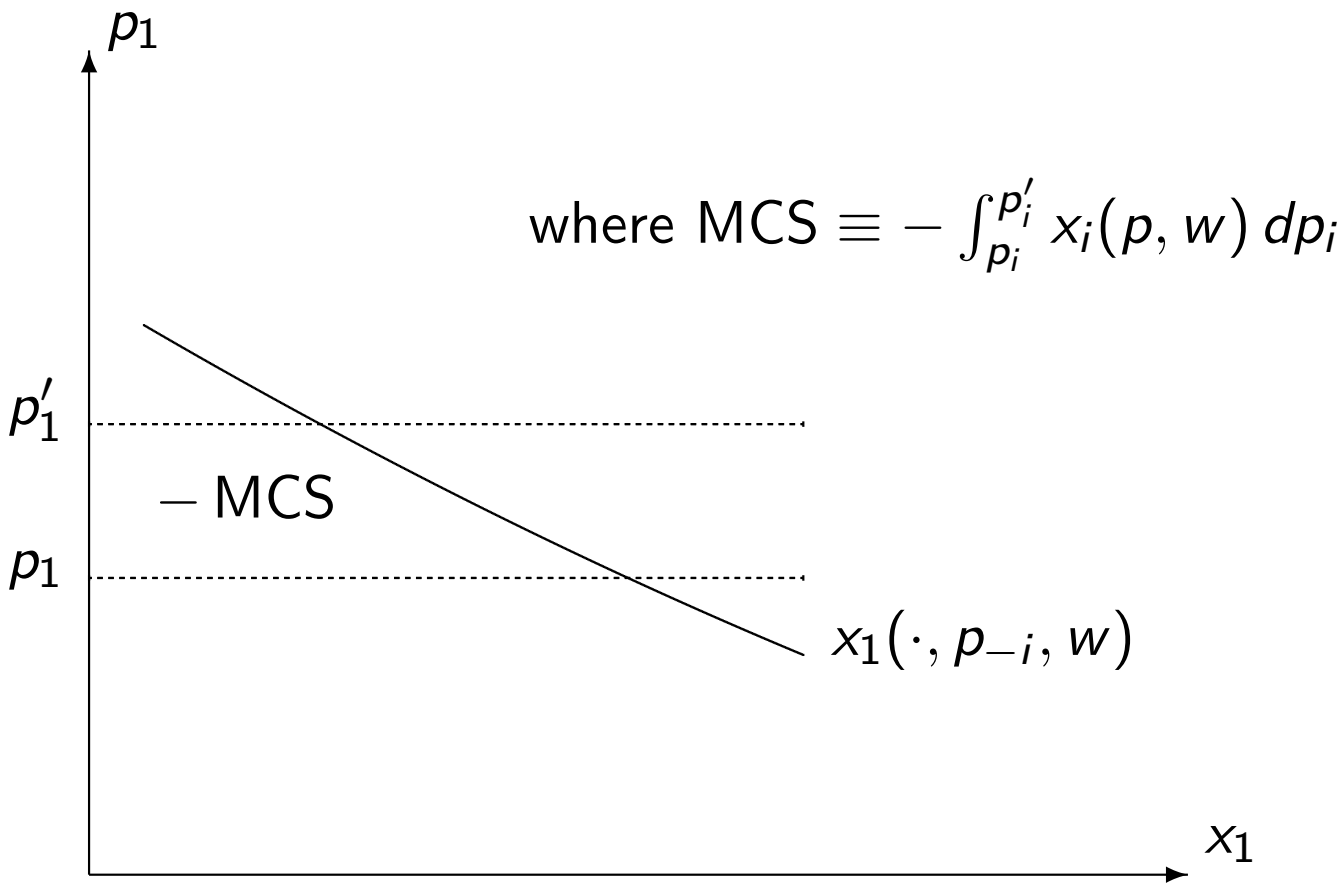
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Welfare and policy evaluation

- In theory, CV or EV can be summed across consumers to evaluate policy impacts
  - If $\sum_i \text{CV}_i > 0$, we can redistribute from "winners" to "losers," making everyone better off under the policy than before
  - If $\sum_i \text{EV}_i < 0$, we can redistribute from "losers" to "winners," making everyone better off than they would be if policy were implemented

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- In reality, identifying winners and losers is difficult
- In reality, widescale redistribution is generally impractical
- Sum-of-CV/EV criterion can cycle (i.e., it can look attractive to enact policy, and then look attractive to cancel it)

Outline

- The welfare impact of price changes
- Price indices
  - Price indices for all goods
  - Price indices for a subset of goods
- Aggregating across consumers
- Optimal taxation

Motivation for price indices

Problem: We generally can't access consumers' Hicksian demand correspondences (or even Marshallian ones)

We can say consumers are better off whenever wealth increases more than prices... but change of what prices?

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1. Ideally we would look at the changing "price" of a "util"
2. Since we can't measure utils, use change in weighted average of goods prices... but with what weights?

The Ideal index

The "price" of a "util" is expenditures divided by utility: $\frac{e(p,\bar{u})}{\bar{u}}$

Definition (ideal index)

$$\text{Ideal Index}(\bar{u}) \equiv \frac{p'_{\text{util}}}{p_{\text{util}}} = \frac{e(p',\bar{u})/\bar{u}}{e(p,\bar{u})/\bar{u}} = \frac{e(p',\bar{u})}{e(p,\bar{u})}.$$

Question: what $\bar{u}$ should we use?

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Question: what $\bar{u}$ should we use? Natural candidates are

- $v(p,w)$; note $e\big(p,v(p,w)\big) = w$, so denominator equals w
- $v(p',w')$; note $e\big(p',v(p',w')\big) = w'$, so numerator equals w'

Ideal index gives change in wealth required to keep utility constant

Weighted average price indices

We can't measure utility and don't know expenditure function $e(\cdot, \bar{u})$, so settle for an index based on weighted average prices

What weights should we use?

Weighted average price indices

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What weights should we use? Natural candidates are

- Quantity x of goods purchased at old prices p
- Quantity x' of goods purchased at new prices p'

The quantities used to calculated weighted average are often called the "basket"

Defining weighted average price indices

Definition (Laspeyres index)

$$\text{Laspeyres Index} \equiv \frac{p' \cdot x}{p \cdot x}$$

Definition (Paasche index)

$$\text{Paasche Index} \equiv \frac{p' \cdot x'}{p \cdot x'}$$

Defining weighted average price indices

Definition (Laspeyres index)

$$\text{Laspeyres Index} \equiv \frac{p' \cdot x}{p \cdot x} = \frac{p' \cdot x}{w} = \frac{p' \cdot x}{e(p, \bar{u})},$$

where $\bar{u} \equiv v(p, w)$.

Definition (Paasche index)

$$\text{Paasche Index} \equiv \frac{p' \cdot x'}{p \cdot x'} = \frac{w'}{p \cdot x'} = \frac{e(p', \bar{u}')}{p \cdot x'},$$

where $\bar{u}' \equiv v(p', w')$.

Bounding the Laspeyres and Paasche indices

Note that since $u(x) = \bar{u}$ and $u(x') = \bar{u}'$, by "revealed preference"

$$p' \cdot x \geq \min_{\xi:\, u(\xi) \geq \bar{u}} p' \cdot \xi = e(p', \bar{u})$$

$$p \cdot x' \geq \min_{\xi:\, u(\xi) \geq \bar{u}'} p \cdot \xi = e(p, \bar{u}')$$

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Thus we get that the Laspeyres index overestimates inflation, while the Paasche index underestimates it:

$$\text{Laspeyres} \equiv \frac{p' \cdot x}{e(p, \bar{u})} \geq \frac{e(p', \bar{u})}{e(p, \bar{u})} \equiv \text{Ideal}(\bar{u})$$

$$\text{Paasche Index} \equiv \frac{e(p', \bar{u}')}{p \cdot x'} \leq \frac{e(p', \bar{u}')}{e(p, \bar{u}')} \equiv \text{Ideal}(\bar{u}')$$

Why the Laspeyres and Paasche indices are not ideal

Deviation of Laspeyres/Paasche indices from Ideal comes from

$$p' \cdot x \geq p' \cdot h(p', \bar{u}) = e(p', \bar{u})$$
$$p \cdot x' \geq p \cdot h(p, \bar{u}') = e(p, \bar{u}')$$

The problem is that

- $p' \cdot x$ doesn't capture consumers' substitution away from x when prices change from p to p'
- $p \cdot x'$ doesn't capture consumers' substitution to x' when prices changed from p to p'

Particular forms of this substitution bias include

- New good bias
- Outlet bias

Price indices for a subset of goods

Suppose we can divide goods into two "groups"

1. Goods E: $\{1, \ldots, k\}$
2. Other goods $\{k+1, \ldots, n\}$

A meaningful price index for E requires that consumers can rank p_E without knowing p_{-E}

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A meaningful price index for E requires that consumers can rank p_E without knowing p_{-E}

For welfare ranking of price vectors for E not to depend on prices for other goods, we must have

$$e(p_E, p_{-E}, \bar{u}) \leq e(p'_E, p_{-E}, \bar{u}) \iff$$
$$e(p_E, p'_{-E}, \bar{u}') \leq e(p'_E, p'_{-E}, \bar{u}')$$

for all p_E, p'_E, p_{-E}, p'_{-E}, $\bar{u}$, and $\bar{u}'$

A "separability" result for prices

Recall

Theorem

Suppose $\succsim$ *on* $X \times Y$ *is represented by* $u(x, y)$*. Then preferences over* X *do not depend on* y *iff there exist functions* $v : X \to \mathbb{R}$ *and* $U : \mathbb{R} \times Y \to \mathbb{R}$ *such that*

1. $U(\cdot, \cdot)$ *is increasing in its first argument, and*
2. $u(x, y) = U\big(v(x), y\big)$ *for all* (x, y)*.*

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1. *$U(\cdot, \cdot)$ is increasing in its first argument, and*
2. *$u(x, y) = U\big(v(x), y\big)$ for all (x, y).*

Theorem

Welfare rankings over p_E do not depend on p_{-E} iff there exist functions $P : \mathbb{R}^k \to \mathbb{R}$ and $\hat{e} : \mathbb{R} \times \mathbb{R}^{n-k} \times \mathbb{R} \to \mathbb{R}$ such that

1. *$\hat{e}(\cdot, \cdot, \cdot)$ is increasing in its first argument, and*
2. *$e(p, \bar{u}) = \hat{e}\big(P(p_E), p_{-E}, \bar{u}\big)$ for all p and $\bar{u}$.*

Price indices for a subset of goods: other result

Results include that

- This separability in e gives that Hicksian demand for goods outside E only depend on p_E through the price index $P(p_E)$
- $P(\cdot)$ is homothetic (i.e., $P(p'_E) \geq P(p_E) \iff P(\lambda p'_E) \geq P(\lambda p_E)$); we can therefore come up with *some* $P(\cdot)$ which is homogeneous of degree one
- Neither of the two separability conditions defined by the theorems on the previous slide imply each other

More detail is in the lecture notes

Outline

- The welfare impact of price changes
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We can't model the individual consumers in an economy

There are typically too many consumers to model explicitly, so we consider a small number (often only one!)

- Valid if groups of consumers have same preferences and wealth
- If consumers are heterogeneous, validity of aggregation depends on
  - Type of analysis conducted
  - Form of heterogeneity

We consider several forms of analysis: under what forms of heterogeneity can we aggregate consumers?

Types of analysis conducted in the face of heterogeneity

We might try to

1. Model aggregate demand using only aggregate wealth
2. Model aggregate demand using wealth and preferences of a single consumer (i.e., a "positive representative consumer")
3. Model aggregate consumer welfare using welfare of a single consumer (i.e., a "normative representative consumer")

Modelling aggregate demand using aggregate wealth I

Question 1

Can we predict aggregate demand knowing only the aggregate wealth and not its distribution across consumers?

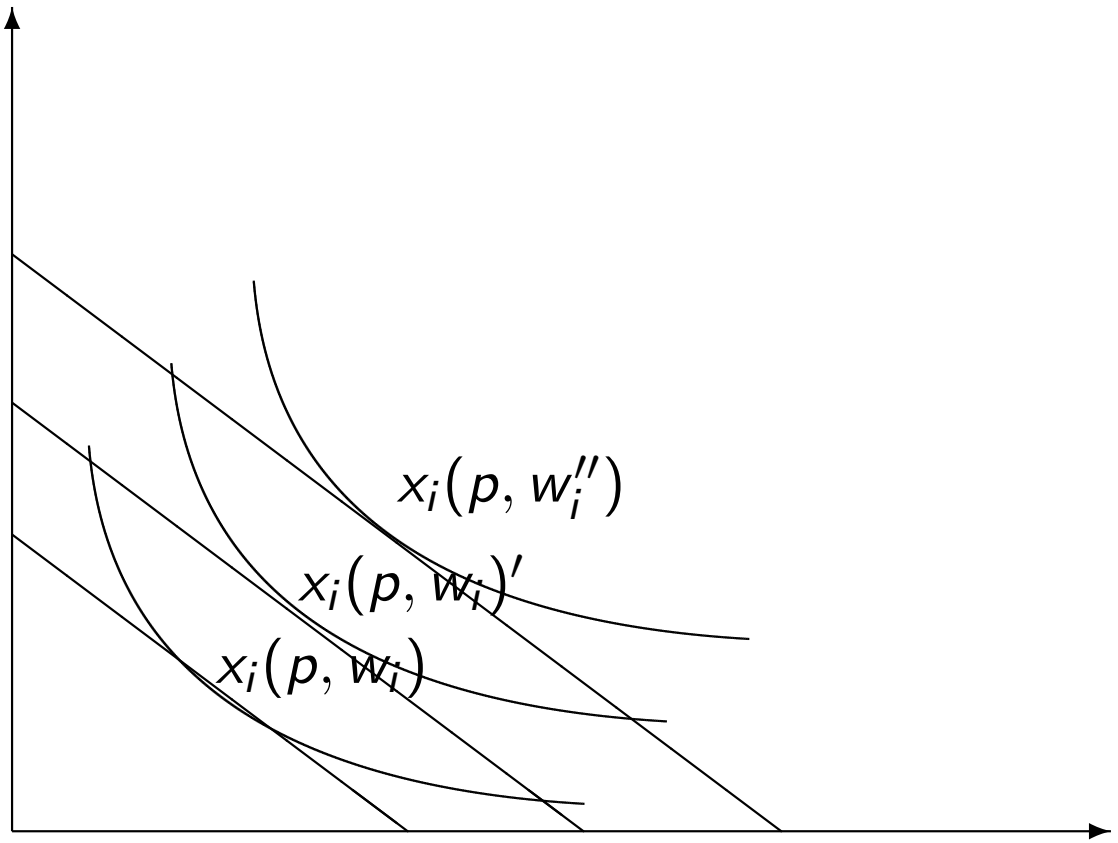
Necessary and sufficient condition: reallocation of wealth never changes total demand; i.e.,

$$\frac{\partial x_i(p, w_i)}{\partial w_i} = \frac{\partial x_j(p, w_j)}{\partial w_j}$$

for all p, i, j, w_i, and w_j

Modelling aggregate demand using aggregate wealth II

- Engle curves must be straight lines, parallel across consumers
- Consumers' indirect utility takes Gorman form:
  $v_i(p, w_i) = a_i(p) + b(p)w_i$



Aggregate demand with positive representative consumer

Question 2

Can aggregate demand be explained as though arising from utility maximization of a single consumer?

Aggregate demand with positive representative consumer

Question 2

Can aggregate demand be explained as though arising from utility maximization of a single consumer?

Answer: Not necessarily

Aggregate welfare with normative representative consumer

Question 3

Assuming there is a positive representative consumer, can her welfare be used as a proxy for some welfare aggregate of individual consumers?

Aggregate welfare with normative representative consumer

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Assuming there is a positive representative consumer, can her welfare be used as a proxy for some welfare aggregate of individual consumers?

Answer: Not necessarily

How does this work for firms?

Looking forward to our discussion of general equilibrium, we can also ask about aggregation across firms

Firms aggregate perfectly (assuming price-taking): given J firms,

- Aggregate supply as if single firm with production set

$$Y = Y_1 + \cdots + Y_J = \left\{ \sum_{j=1}^{J} y_j : y_j \in Y_j \text{ for each firm } j \right\}$$

- Profit function $\pi(p) = \sum_j \pi_j(p)$

Firms can aggregate because they have no wealth effects

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How should consumption be taxed I

Suppose we can impose taxes t in order to fund some spending T
What taxes should we impose? Several ways to approach this

1. Maximize $v(p+t, w)$ such that $t \cdot x(p+t, w) \geq T$
2. Minimize $e(p+t, \bar{u})$ such that $t \cdot h(p+t, \bar{u}) \geq T$

Following the second approach gives Lagrangian

$$\mathcal{L} = -e(p+t, \bar{u}) + \lambda\big(t \cdot h(p+t, \bar{u}) - T\big)$$

And FOC

$$\nabla_p e(p+t^*, \bar{u}) = \lambda h(p+t^*, \bar{u}) + \lambda\big[\nabla_p h(p+t^*, \bar{u})\big]\, t^*$$

How should consumption be taxed II

$$\underbrace{\nabla_p e(p + t^*, \bar{u})}_{h(p+t^*,\bar{u})} - \lambda h(p + t^*, \bar{u}) = \lambda \left[\nabla_p h(p + t^*, \bar{u})\right] t$$

$$\frac{1 - \lambda}{\lambda} h(p + t^*, \bar{u}) = \left[\nabla_p h(p + t^*, \bar{u})\right] t^*$$

$$\frac{1 - \lambda}{\lambda} \left[\nabla_p h(p + t^*, \bar{u})\right]^{-1} h(p + t^*, \bar{u}) = t^*$$

This is a generally a difficult system to solve

The no-cross-elasticity case

If $\frac{\partial h_i}{\partial p_j} = 0$ for $i \neq j$, we can solve on a tax-by-tax basis:

$$\lambda t_i^* \frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} = \underbrace{\frac{\partial e(p + t^*, \bar{u})}{\partial p_i}}_{= h_i(p + t^*, \bar{u})} - \lambda h_i(p + t^*, \bar{u})$$

$$\lambda t_i^* \frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} = (1 - \lambda) h_i(p + t^*, \bar{u})$$

$$t_i^* = \frac{1 - \lambda}{\lambda} h_i(p + t^*, \bar{u}) \left[\frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} \right]^{-1}$$

$$\frac{t_i^*}{p_i} = \frac{1 - \lambda}{\lambda} \left[\frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} \frac{p_i}{h_i(p + t^*, \bar{u})} \right]^{-1}$$

So optimal tax rates are proportional to the inverse of the elasticity of Hicksian demand